SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation?

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Abstract

Dynamic manipulation requires a robot to coordinate contact with a scene that changes between observation and execution. We study which history encodings, future representations, and world–action connections help action generation. Our guiding hypothesis is that prediction should preserve target motion and task-relevant context at the granularity required by interaction. A fifteen-condition auxiliary-supervision study on fourteen simulated tasks gives a highest success rate of 64.13% on 1,260 paired scenes, compared with 62.78% without future supervision, with no clear overall advantage. Dense 2-D displacement supervision reaches 59.21% and is worse under the reported multiplicity correction. On 1,400 scenes with current GT target pose, no visual-history encoder exceeds the current-image reference (53.43%); a GT-motion reference reaches 58.86%. Replacing past object poses with the current pose at inference reduces this frozen reference to 0.64%, revealing dependence on correct motion history. Under CNN/reconstruction, a trajectory-supervised configuration selected on the 1,400-scene development pool reaches 66.79% at seed 0 and 61.21% at seed 1. The matched no-future and scene-flow/pose controls reach 57.86% and 61.50% at seed 0; their repeats remain pending. Of the 74 additional seed-0 successes over the latter control, 72 are on phase passage. The main screens use one training seed and privileged target information. Their conditional results guide the remaining backbone, online information-flow, pre-training, and external evaluations needed to determine the final SE(3)-WAM instance.

1. Introduction

A moving object can leave its observed position before a robot reaches it. The same problem arises when a receptacle moves during release or an obstacle changes the available approach corridor. Dynamic manipulation therefore requires spatial accuracy at the right time. Faster inference and more responsive execution reduce observation age [33], while temporal observations help distinguish motions that look similar in a single image [9, 23]. Together, these demands motivate a practical question: *what should a robot remember and predict so that its action remains useful when interaction occurs?*

The answer depends on the role of information in the task. Target translation and rotation determine where and how to make contact. A moving occluder or obstacle matters through its effect on visibility, reachability, or collision. Static geometry can remain essential even though it has no motion to predict. This suggests organizing scene evolution by interaction needs: preserve fine target information when contact requires it, and represent surrounding changes at sufficient resolution to support the decision. Both a fixed target/context decomposition and a shared representation with task-dependent readout can implement this idea. Their relative value is an empirical question.

World–action models provide a framework for learning scene evolution alongside robot actions [36, 37]. However, a future objective changes several things at once: the supervised variables, the gradients shaping shared features, and sometimes the computation used before acting. The action may benefit from auxiliary supervision even when no forecast is generated at deployment [39]. Conversely, a geometrically accurate forecast may add no useful information to an already strong state-conditioned policy. Prediction error, action improvement, and online forecast use must therefore be measured separately.

We organize the SE(3)-WAM Study around three connected questions. **Q1** asks how a finite history should encode motion. **Q2** asks which future variables, regions, and resolutions support action. **Q3** asks how world and action computation should share parameters and exchange information. Each study matches its own reference; history, future, and combination studies use distinct recipes. Cross-comparisons revisit competitive choices before robot-data pretraining.

The completed comparisons give different outcomes across recipes. Structured GT motion improves the history reference, and a frozen-policy intervention reveals dependence on its correctness. Visual-history decodability and

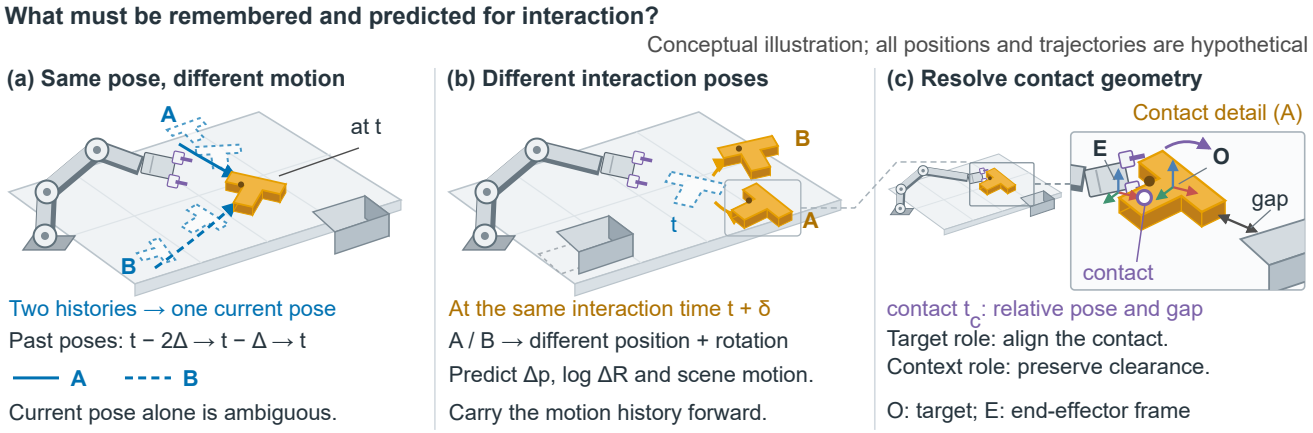


Figure 1. What information matters at interaction time? (a) Two histories meet at the same current target pose. (b) They lead to different poses at the same future time (solid A, dashed B). (c) A contact detail for A shows end-effector/target alignment and context clearance. The same asymmetric target and robot connect the panels. All trajectories are conceptual illustrations of the research question.

action sensitivity do not translate into an overall success-rate advantage. Among fifteen future-supervision conditions on 1,260 paired scenes, sparse trajectories give the highest point estimates but no clear overall gain; dense 2-D displacement decreases success. A twelve-condition combination screen yields a candidate whose gains over matched controls are concentrated on phase passage. An adapted DreamVLA system remains stronger under its own recipe. These results motivate testing how information affects action, rather than treating added history or future targets as established benefits.

The current evidence supports two contributions:

- A dynamic-manipulation study protocol that separates task outcomes, available information, training supervision, and execution conditions.
- Controlled history and future-supervision studies that distinguish readable history, action sensitivity, and task benefit, including a frozen-policy motion-history intervention and fifteen future-supervision conditions.

The supplement details the measured configurations and the remaining design, pretraining, and transfer protocols.

2. Related Work

Dynamic manipulation and physical time. Motion-aware grasping connects prediction to reachability and interception [1]. DOMINO/PUMA couples a dynamic benchmark with historical flow and object-future supervision [9]; DynamicManip expands static demonstrations into dynamic data [17]. DynamicVLA emphasizes continuous inference and action streaming [33], while DynamicWAM conditions world–action learning on motion [23]. Temporal feature prediction [42], AHEAD's predictive latents [27], and ForeTime-VLA's future-token and event-time supervision [24] offer different ways to anticipate interaction. RTC addresses continuity under delayed chunk generation [3]; Chronos Policy explicitly represents observation and execution times [29]. These works motivate evaluating prediction together with action age, rather than identifying a simulator control interval with real-time operation.

History and object-centered information. TraceVLA encodes temporal traces as visual prompts [45]; HiF-VLA organizes past and future dynamics through motion representations [18]. PACT-WAM allocates fewer spatial tokens to older observations [21], and Long-WAM studies long context together with causal video pretraining [13]. Thus a history encoder's value depends on both temporal support and the training that makes that information usable. PointWorld explicitly assumes a static initial scene when neither prior frames nor velocities are supplied [12]. Externally moving targets make this initial-state information a central factor in Q1. DOCIR separates manipulated objects, obstacles, and the robot [8]; OA-WAM makes objects addressable within world–action modeling [22]. Our Q1 comparisons distinguish current localization, privileged motion, and visual history before comparing encodings. Role assignment is an experimental factor, not a presumed optimum.

Which future should a policy predict? SE3-Nets models scene motion through rigid transforms [4]; SE3-Pose-Nets uses structured pose dynamics for control [5]. DSR-Net accumulates a volumetric object representation over interactions [35]; PhysMani learns 3-D world representations for manipulation [40]. ATM learns point trajectories from video [31]; Im2Flow2Act uses generated object flow as a policy interface [34]. AffordTrajDP propagates contact affordances with object pose [32], and FlowWAM uses flow in

policy and world-model modes [6]. These interfaces differ in whether they describe desired motion, action-conditioned consequences, or externally moving objects. DreamVLA and DreamWAM add complementary visual, geometric, and semantic targets [38, 41]; DC-WAM instead emphasizes dynamics within RGB supervision [14]. DELE-w0.5 uses compact future latents [16]. Q2 places related target families under one action path to measure their conditional training value; its pose output does not imply a new rigid-motion parameterization.

World–action coupling and experimental attribution. Prediction enters control through different interfaces. PointWorld predicts action-conditioned 3-D point flows for candidate-action evaluation [12]; NWM simulates and ranks navigation trajectories [2]. CoT-VLA generates a visual goal before its action sequence [43], while VPP conditions actions on predictive features [11]. DreamZero jointly models world and action [36]; Fast-WAM separates training benefits from test-time imagination [39]. Our reference D uses future supervision without a forecast-to-action edge. Q3 changes this access and generation order under controlled inputs. Wang et al.'s OpenWAM supplies the Alpha foundation for our adapted baseline [30]; Yu et al.'s OpenWAM studies composable world–action programs [37]. ACT, Diffusion Policy, and $\pi_{0.5}$ supply system references [7, 25, 44]; DROID supplies the planned real-robot pretraining data [15]. Native system comparisons and matched design contrasts answer different questions and are reported separately.

3. Problem and Evaluation Protocol

Interaction in physical time. Let $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{i \le t}$ denote the available finite history of observations, robot state, and timestamps. A policy produces an action chunk $A_t = (a_{t,0}, \ldots, a_{t,H_a-1})$. If the latest image was captured at τ_t^s and the first command is applied after delay ℓ_t, its application times are

$$\tau_{t,j}^{\text{apply}} = \tau_t^s + \ell_t + j\Delta_a. \tag{1}$$

The relevant future is defined relative to these physical times and the task's contact event. Forecast horizons, action intervals, and computation time are different quantities. In the completed representation screen, labels use eight future samples from 50 to 400 ms, while evaluation uses synchronous policy queries and executes one action per replan. Native command delay is 50 ms for T5 and zero for the other tasks; measured computation time is not injected.

Tasks and data. The reported experiments use the migrated DB0 v0.6 data and the fourteen-task evaluation suite. The suite covers moving-target grasping, orientation-sensitive grasping, physical occlusion, moving-receptacle placement, phase-constrained passage, free rolling/sliding motion, and post-impact adaptation. The simulation infrastructure uses MuJoCo and robosuite [28, 46]. Task success is determined by each native evaluator's physical outcome; all fourteen tasks receive equal weight. The common demonstration pool contains 11,200 trajectories, 800 per task. Appendix A specifies the task and split identities; Supplementary Fig. S1 shows recorded task observations.

Information tracks. The future-supervision screen uses a current fixed-camera RGB image, exact depth/camera geometry, an eight-sample robot-state history, and privileged current target pose plus seven intervals of target kinematics. We call this the *object-motion information track*. It isolates future objectives when substantial target motion information is already available. The current-pose history study removes privileged interval kinematics and asks whether visual history supplies useful additional information. Both tracks still use privileged current target pose. A deployable visual system must estimate such quantities from allowed sensors and is evaluated separately. Current-only RGB with robot-state history is denoted H0V; it retains memory.

Evaluation partitions. The fixed140 development set contains ten scenes per task. The disjoint 1,260-scene complement contains ninety per task, giving 1,400 scenes together. The complement is independent of fixed140 membership but belongs to the same validation distribution and has now informed development. The combined 1,400-scene score includes development scenes and is used for combination selection; it is not a new out-of-distribution test. We preserve these identities in every table. Checkpoints in the completed future screen use the fixed fourth-epoch endpoint and one fresh training seed. Paired comparisons share scenes and specified rollout seeds; scenario uncertainty does not include between-training-seed variation.

Outcomes and comparison scope. For $K = 14$ tasks with n_k episodes each, the primary score is

$$\mathrm{SR}(m) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}, \qquad y_{mki} \in \{0,1\}. \tag{2}$$

Equal per-task counts make pooled and task-balanced SR identical here. We report counts as well as percentages. Paired differences use the same episodes and the uncertainty procedure attached to the source analysis. The completed future comparison corrects its twenty-seven overall contrasts as one family. Per-task findings are exploratory.

Matched internal studies share observation fields, training exposure, action interface, and execution settings. The external-method adaptation table instead compares complete

systems with different pretraining, training budgets, and decoders. It records a useful performance reference without identifying a particular architecture's effect. Contact-time and relative-pose errors, safety outcomes, and measured action age are reserved for the mechanism and deployment studies where their measurement interfaces are available. Missing readouts are not assigned zero. Representation-specific errors retain their native units; a latent loss is not ranked numerically against a metric pose error.

4. The SE(3)-WAM Design Study

The study represents a design by (H, Z, B, G): a history encoder H, future targets Z, backbone B, and information-flow program G. Figure 2 instantiates these roles in reference D. Completed studies examine H and Z under their respective references; planned studies vary B and G and revisit retained alternatives. A single component's point-estimate ranking does not determine the final composition.

4.1. Q1: Information Access and History Encoding

The encoder maps timestamped history into conditioning tokens, $X_t = H(\mathcal{H}_t)$. Its design includes the physical window, temporal spacing, visibility, and motion coordinates. Current RGB, stacked RGB windows, codec motion vectors, optical flow, latent history, and per-frame query compression provide different descriptions of the same recent evolution. Encoding and information access are separate factors: supplying a ground-truth velocity changes what is known, while reorganizing the same visual samples changes how it is represented.

The current-pose study holds current target pose and eight robot-state samples fixed, removes target-history kinematics, and compares eight visual-history conditions plus a freshly trained GT-history reference. The object-motion track instead supplies seven intervals of target kinematics. This separation tests the conditional value of visual history given target localization. A frozen-policy intervention replaces past object poses with the current pose and recomputes kinematics, testing dependence on correct motion history. Appendix C gives the protocol and planned visibility diagnostics.

4.2. Q2: Predicting Interaction-Relevant Variables

Q2 separates *scope* (target, context, or both), *variables* (pose, motion, visual features), and *granularity* (spatial and temporal allocation). The completed comparison includes no future target, scene-flow only, target-pose only, their combination, alternative scene/target descriptions, and additional future supervision. Figure 3 gives the central contrasts; Appendix D provides all fifteen conditions. Scene flow also covers target regions, so the scene-flow-only condition is not a context-only control. Pure role separation and allocation require additional matched comparisons.

For the rigid-target condition, let (p_t, R_t) be the current target pose in the fixed metric frame. At physical offset $\delta_h = 0.05h$ seconds, $h = 1, \ldots, 8$, the supervised variables are

$$d_h = p_{t+\delta_h} - p_t, \qquad \phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee. \quad (3)$$

Predicted poses reconstruct as $\hat{p}_{t+\delta_h} = p_t + \hat{d}_h$ and $\hat{R}_{t+\delta_h} = \mathrm{Exp}(\hat{\phi}_h^\wedge) R_t$. Translation and left-relative rotation are parameterized separately. The pose readout pools sixteen meta tokens and predicts eight cumulative transforms. Its valid-target loss combines squared translation error with 0.1 times squared rotation geodesic error. This geometric parameterization does not impose equivariance on the network.

Scene-flow supervision describes metric cumulative point displacement. Alternatives describe dense 2-D displacement, codec block motion, sparse scene trajectories with visibility, or future video latents. Target alternatives use pooled DINO features or target trajectories. The added-supervision conditions predict depth, DINO features, dynamic-region RGB, or their combination. Their labels have different units, support, and dimensionality. Replacing scene flow with dense 2-D motion consequently tests a representation package, not dimensionality alone. Likewise, pose plus sparse tracks is not by itself a matched proof that fine-target/coarse-context allocation is optimal.

The completed study extracts future targets along demonstration trajectories and uses them as auxiliary supervision. Validity masks describe geometric support and visibility, not compatibility with a different action sequence. Future labels and validity masks remain in their supervision branches; neither is an action input. Deterministic readouts and generated variables have distinct interfaces. To compare uniform and heterogeneous allocation directly, the remaining study must hold role information, temporal support, and an applicable token/compute budget common, then vary the allocation within a compatible representation. A target mask used to construct a training label does not establish deployable target assignment.

4.3. Reference Training and Action Generation

The common Q2 reference uses a four-layer transformer of width 768 with eight attention heads. A frozen DINOv3 ViT-L/16 encoder [26] provides 196 tokens from a 224×224 current image. The D backbone shares attention projections and uses separate feed-forward networks for world/meta, flow, and action streams. Action attends to meta and action tokens; it does not attend to predicted flow or flow hidden states. Future objectives can still shape shared trainable features through their gradients.

Action generation uses flow matching [19]. With normalized action target a_1, source $a_0 \sim \mathcal{N}(0, I)$, and interpolation

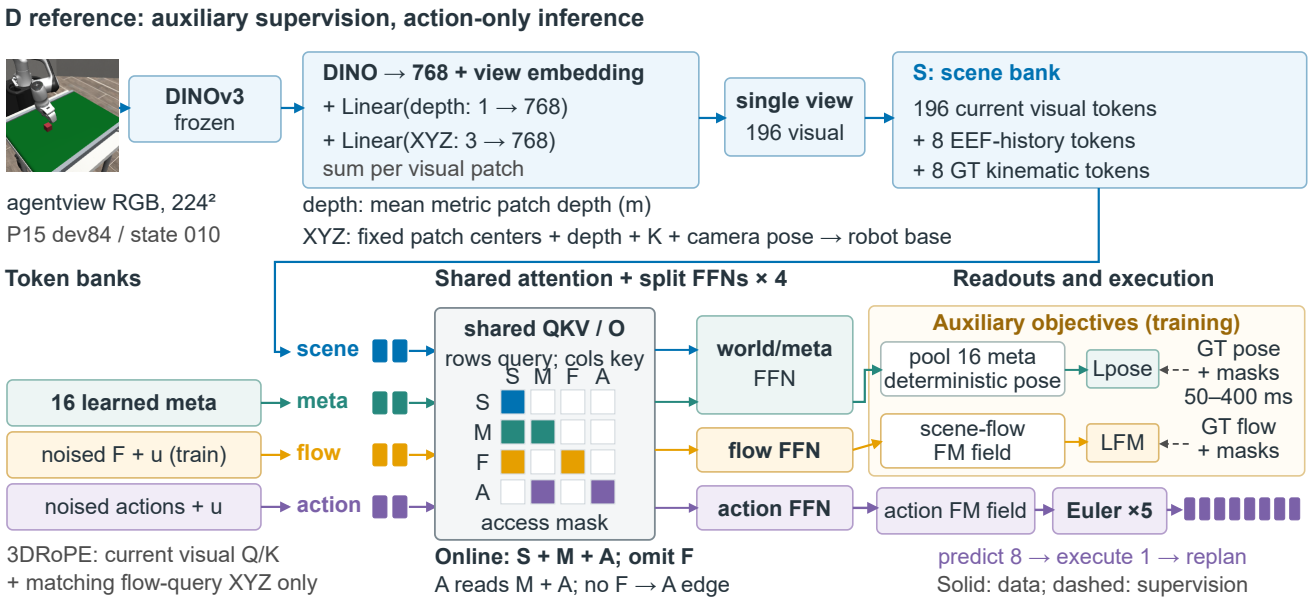


Figure 2. Implemented reference D, with P03's pose and scene-flow targets shown. RGB with depth/camera geometry, robot history, and privileged target state form the scene bank; learned meta tokens are separate inputs. The mask gives query-to-key access for scene (S), meta (M), flow (F), and action (A). Within each of four layers, the streams share attention projections and use stream-specific FFNs. Pose is a deterministic readout; scene flow and action use flow-matching fields. Future labels and masks provide training supervision (dashed). Deployment solves only action with five Euler steps, executes the first E7 command, and replans. Appendix B gives embedding and target definitions. The RGB inset is a real benchmark observation; the diagram describes D's computation.

time $u \in [0,1]$,

$$a_u = (1-u)a_0 + ua_1, \quad \mathcal{L}_A = \mathbb{E}\|v_\theta(a_u, u, X_t) - (a_1 - a_0)\|^2. \tag{4}$$

The complete objective adds only the enabled future losses, $\mathcal{L} = \mathcal{L}_A + \sum_{r \in Z} \lambda_r \mathcal{L}_r$. Masks restrict each loss to its valid support. A missing head contributes no measurement; raw total losses with different terms are not compared as one quality metric. Appendix B specifies the reference source distributions, coordinates, and implementation settings.

The action chunk contains eight seven-dimensional commands: translation increment, left-relative rotation vector, and a gripper command. At deployment, five Euler steps generate the action chunk, its first command is executed, and the policy replans. Conditioning caching is enabled; temporal and velocity caching are disabled. The training/evaluation translation bounds are respectively 0.05/0.10 m per axis. Only the action stream is solved online in the completed Q2 screen.

4.4. Q3: Backbone and Information Flow

The next backbone comparison holds selected H/Z and the reference action-only deployment program G_0 fixed. It compares D's stream-specific feed-forward organization with an adapted E world/action parameter organization. The planned comparison trains both from scratch; the active-backbone parameter difference is targeted within 5%. The matched backbone results are **[pending]**.

The subsequent seven-condition information-flow study holds $H/Z/B$ fixed: R0 removes the studied future loss; R1 retains it without cross-stream edges; R2 allows future-to-action access; R3 allows action-to-future access; R4 allows both; R5 generates future then action; R6 generates action then future. R1 also compares action-only and joint deployment of the same checkpoint. These contrasts distinguish supervision, information access, and generation programs. The sequential conditions use clean ground-truth first-stage values for second-stage training and predicted values at deployment, so their difference includes teacher forcing and its deployment mismatch.

The current generative interface supports scene flow; sparse tracks or video latents require a corresponding generated-state definition. Five joint calls and five-plus-five sequential calls match per-modality steps but differ in total compute, which is measured separately. Appendix E specifies the interface, cost comparison, and online forecast interventions.

5. Experiments

We distinguish three recipes. The study's Alpha loss regresses cumulative 3-D displacement magnitudes. The future screen uses current RGB, DINO-L, GT target kinematics, Alpha off, and four-epoch cosine decay. The history screen

Table 1. System references on the 1,260-scene complement (fourteen tasks, ninety scenes each). Training seed is 0. Methods retain their adapted input, initialization, and execution pipelines; budgets and fixed140 counts are in Appendix B. Scores compare complete systems, not matched architectures. The lower block receives GT target pose and interval kinematics.

System	Initialization / study condition	Successes /1,260	SR (%)
Adapted published systems			
ACT [44]	From scratch	424	33.65
Diffusion Policy [7]	From scratch	432	34.29
$\pi_{0.5}$ [25]	Official `pi05_base`	751	59.60
DynamicVLA [33]	Adapted native pipeline	363	28.81
OpenWAM-Alpha [30]	Official Alpha foundation	708	56.19
DynamicWAM [23]	Official full initialization	670	53.17
DreamVLA [41]: 20 epochs	Independent 20-epoch training schedule	1,000	79.37
40 epochs	Independent 40-epoch training schedule	1,003	79.60
Controlled-study systems with privileged target information			
P00: no future supervision	Current image; frozen DINO-L	791	62.78
P03: scene flow + pose	Current image; frozen DINO-L	786	62.38
I: scene tracks + pose	Full visual window; CNN; target RGB reconstruction	839	66.59

uses current GT pose, DINO-L, and pose/flow/Alpha supervision, with one-epoch cosine decay then a constant learning rate. Combinations use full T8 RGB, GT kinematics, DINO-S or CNN, Alpha off, and four-epoch cosine decay. All retain robot history (Appendices B–D).

5.1. System References and Training Conditions

Table 1 places the study against adapted action and world–action models on the same fourteen-task evaluation scenes. The two DreamVLA schedules reach 1,000/1,260 and 1,003/1,260 successes. The study's combination candidate I reaches 839/1,260, and the no-future reference reaches 791/1,260. These are system-level comparisons: adapted methods differ in initialization, training exposure, observation processing, and action execution. Candidate I and the no-future reference also have privileged target information. The scores do not isolate the effect of future representation.

Within the fifteen-condition future screen, all task networks start fresh with training seed 0 and run four epochs, 40,104 updates, and 641,664 sampled chunks at global batch 16. They use the fixed fourth-epoch checkpoint. Training is distributed over eight or sixteen devices with the same global batch; topology histories are retained in Appendix B. A single trained endpoint per condition supports paired scene comparisons but does not measure training-seed robustness. Current pose, motion history, action objective, and one-step receding-horizon execution are held common across this screen.

5.2. Q2: Which Future Supervision Helps?

Adding scene-flow and target-pose objectives. Figure 3 places the core contrasts in the full comparison. The no-future reference P00 succeeds in 791 scenes. Scene-flow P01, target-pose P02, and their combination P03 succeed in 782, 789, and 786 scenes, respectively. The combined objective does not show an overall improvement over either component or the reference in this setting. This result concerns auxiliary training with privileged motion input; online forecast consumption is a separate contrast.

Changing the predicted variables. Sparse scene tracks with target pose (P07) reach 807/1,260, and scene flow with target tracks (P09) reaches 808/1,260. Relative to P00, P07 gains 85 scenes and loses 69; P09 gains 91 and loses 74. Their raw two-sided exact McNemar p values are 0.227 and 0.213. The small net gains do not establish a reliable overall advantage. Non-rejection also does not establish equivalence: P12, for example, has the same total as P00 but exchanges 73 successes and 73 failures.

Dense 2-D future displacement with pose (P05) reaches 746/1,260, a decrease of 3.57 percentage points from P00 and 3.17 points from P03. The corresponding Bonferroni-adjusted p values over twenty-seven overall contrasts are 0.00844 and 0.01877. The comparison supports an adverse effect for this representation condition under the reported recipe. It does not isolate a universal 2-D versus 3-D ordering: P07 and P09 also use 2-D trajectories, with different support and target definitions.

Task variation and failure evidence. All fifteen conditions obtain zero successes on rigid-motion receptacle placement (`t2_se3`). On full constrained passage (`t3_full`), P00 and P06 obtain 3/90 and 4/90, with zero for the remaining conditions. Prediction changes have therefore not removed these shared task failures. P05's loss relative to P03 includes 27 fewer successes on rolling-sphere capture; its

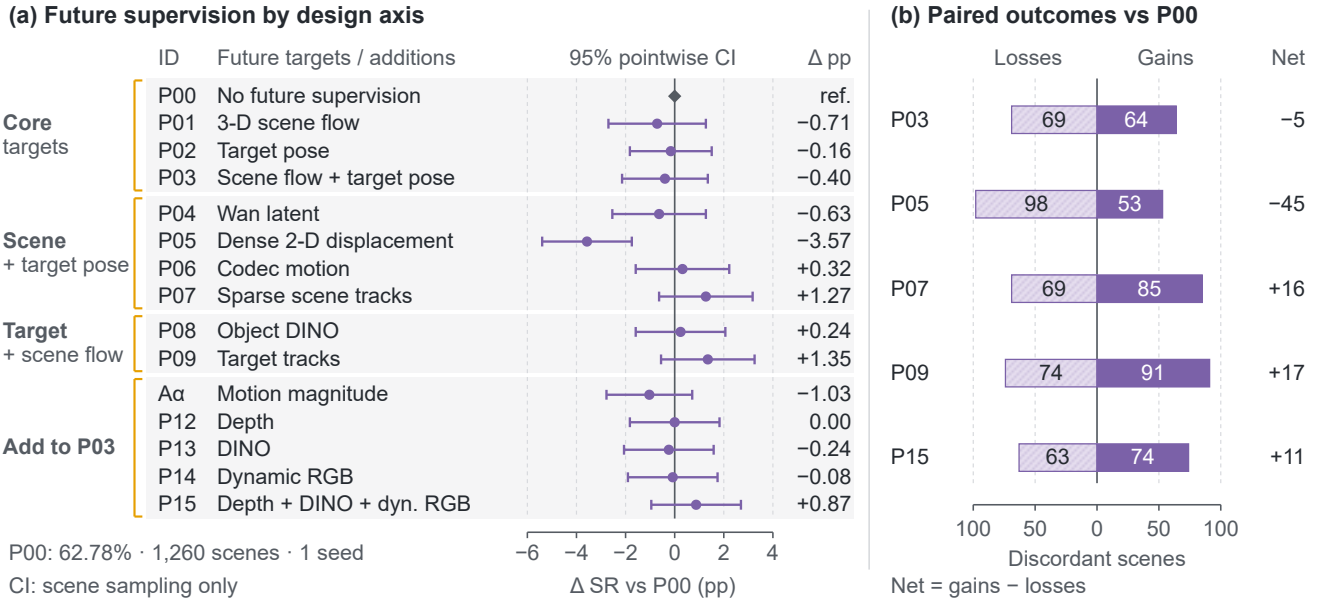


Figure 3. Future supervision at the fourth-epoch endpoint: one training seed, 1,260 paired scenes per condition (fourteen tasks, ninety each). (a) All fifteen conditions, grouped by the component changed; points show success-rate differences from P00 (62.78%), with 95% pointwise intervals from task-stratified paired bootstrap. (b) Selected contrasts expose scene-level changes: gain means P00 fails and the candidate succeeds; loss means the reverse. Net change is gain minus loss. P05 is adverse after the 27-contrast correction; P07/P09 do not establish an overall gain. Intervals condition on the trained checkpoints and exclude training-seed variability. The separate fixed140 partition is in Appendix D.

native `tray_contact` failures increase from 40 to 67 in that task. P09 gains on moving placement and replanning but loses on orientation-sensitive grasping. These observed failure codes and counts identify where to investigate; they do not identify a representation mechanism without an aligned intervention.

Development-set ordering. P01 has the highest fixed140 point estimate, while P09 has the highest 1,260-scene point estimate. The development-set ranking does not persist across the remaining validation scenes. Appendix D reports all conditions, paired gain/loss counts, and per-task results. The evidence favors retaining competitive alternatives for joint comparison instead of treating the first ranking as a final design principle.

5.3. Combining Future Targets with Visual Encoding

The twelve-condition combination study crosses P07/P09/P15 with DINO-S or a CNN encoder and with or without current-target RGB reconstruction. It uses a full visual window and privileged target-motion information. The predefined selection score is success over all 1,400 development/evaluation scenes. Candidate I, P07 with CNN and reconstruction, obtains 935/1,400 (66.79%), including 839/1,260 (66.59%) on the complement, and is selected for matched controls (Table 2).

Table 2. Future–encoder combinations at seed 0: successes /1,400 (the development selection score, including fixed140). Rec. denotes current-target RGB reconstruction. All conditions receive a full visual window and GT target motion; DINO-S is frozen and the CNN trains from scratch. † marks selection by the development score. Both matched controls are complete. Dashes in the control rows denote unrun combinations. Appendix D gives task counts and paired outcomes.

	DINO-S		CNN	
Future supervision	No rec.	+rec.	No rec.	+rec.
P15: P03 + three targets	873	836	875	867
P07: scene tracks + pose	840	849	896	935†
P09: flow + target tracks	820	830	829	870
Matched controls for the selected configuration				
P00: no future	—	—	—	810
P03: flow + pose	—	—	—	861

Its 39-scene net gain over the same P07/CNN condition without reconstruction is concentrated: phase passage gains 44 scenes and orientation-sensitive grasping gains 17, while translational placement loses 10 and XYZ grasping loses 7. Both rigid-motion placement and full passage remain at zero for all twelve conditions. Thus the selection is useful

Table 3. Single-seed history comparison with current GT target pose and robot history. GT motion adds privileged target history to current RGB. SR uses the 1,400-scene development total; Δ and its pointwise 95% scene-bootstrap interval are relative to current image, in percentage points. Appendix C gives the protocol, partition counts, and training differences.

History input	SR (%)	Δ	95% CI (pp)
Current image	53.43	0.00	—
Window	47.29	-6.14	$[-8.43, -3.79]$
Codec-MV	52.36	-1.07	$[-3.36, 1.21]$
Flow	51.43	-2.00	$[-4.43, 0.43]$
Flow12D	51.64	-1.79	$[-4.21, 0.64]$
PACT-Age	45.29	-8.14	$[-10.64, -5.64]$
Wan	50.71	-2.71	$[-5.29, -0.14]$
PACT-Uniform	47.86	-5.57	$[-8.00, -3.14]$
GT motion	58.86	$+5.43$	$[3.07, 7.79]$

for subsequent experimentation, but its overall score does not imply uniform task improvement. The matched controls retain the same CNN, visual window, and reconstruction. No-future F obtains 810/1,400 (57.86%), and flow/pose E obtains 861/1,400 (61.50%). Against F, I gains 189 paired scenes and loses 64: a net 125 scenes, or 8.93 percentage points (95% pointwise interval: 7.00–10.86). Against E, I gains 149 and loses 75: 5.29 points (interval: 3.50–7.07). Phase passage contributes 72 net successes in each comparison; the other thirteen tasks together contribute only two against E. The result supports a task-concentrated benefit under this recipe. I was selected from twelve candidates on this pool. A recipe follow-up reproduces its seed-0 score but obtains 857/1,400 at seed 1 (Appendix H); matched E/F repeats and unused-scene confirmation remain **[pending]**.

5.4. Q1: History Information and Control Utility

With current GT target pose and robot history retained, Current obtains 748/1,400 (53.43%); none of seven visual-history encodings exceeds its point estimate (Table 3). Adding GT motion gives 824/1,400 (58.86%), a 5.43-point gain, despite 83 fewer successes on phase passage. These single-seed comparisons have differing device histories (Appendix C).

Window lowers the frozen Meta readout's 350 ms past-position error from 10.89 to 8.46 mm, yet loses 6.14 success-rate points. Past-image perturbations change all seven history models' actions while leaving both current-image references invariant. Thus decodability and action sensitivity do not by themselves establish task benefit.

In the frozen GT-motion endpoint, replacing past poses with the current pose and recomputing kinematics reduces success from 824 to 9/1,400: a 58.21-point drop (95% paired interval 56.50–59.86). All 210 probe-window first actions change, with mean translation change 18.40 mm; identity intervention preserves actions and reproduces all 140 control success/failure labels (Appendix C). This frozen-policy mismatch effect differs from the 5.43-point training-comparison gain. Together, these results support structured motion history in this recipe, without establishing sufficiency or a universal encoder ranking.

5.5. Execution and Learning Diagnostics

Future-screen policy calls average 177.071–184.212 ms on the same device, with 1.571–1.617 GiB peak allocated memory. Appendix F reports p95 times, timing boundaries, and the single-round protocol. These small differences do not support an efficiency ranking; the simulation's 50 ms action interval does not establish 20 Hz wall-clock deployment.

The full-passage diagnostic finds release supervision only at h_7, not the executed h_0, in all 800 demonstrations. Short training and first-action weighting do not establish a consistent remedy (Appendix G). A frozen gripper classifier improves offline NLL but passes only 9/11 predefined regression cases; its new-scene evaluation is therefore not run (Appendix H). Supervision and execution must be checked before attributing difficult-task failure to future representation.

5.6. Remaining Studies and Final Model Selection

The planned D/E and R0–R6 studies revisit retained representations and backbones before multi-seed, unused-scene confirmation (Appendix E). The selected design then undergoes recipe and scale comparisons on DROID, with a separately validated object-pose annotation pipeline. The resulting robot-data base independently initializes simulation and real-robot post-training. Single-arm DOMINO, static compatibility, and real conveyor manipulation assess transfer; measured-delay tests assess contact, action age, and end-to-end cost (Appendix F). These stages remain **[pending]**.

6. Discussion and Conclusion

Structured object-motion history benefits the Q1 reference, and a frozen-policy intervention confirms dependence on its correctness. Visual-history readouts and action sensitivity alone do not establish control utility. Future supervision has conditional value: sparse trajectories give modest point-estimate gains, while dense 2-D displacement is adverse. With CNN/reconstruction, the trajectory condition exceeds both seed-0 controls; its flow/pose advantage concentrates on phase passage. Its score falls from 66.79% to 61.21% at seed 1; matched-control repeats remain pending.

The main screens cover fourteen simulated tasks with privileged target information, synchronous execution, and one training seed. Granularity, online foresight, design stability, and robot transfer require further tests.

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Supplementary Material

This supplement documents the measured experiments, implementation, and remaining study protocols. Sections A–D give the data, configurations, and results; E–F specify pending comparisons; G presents diagnostics and offline predictions; H reports the completed recipe and frozen-decoder studies.

Appendix	Contents	Main
A	Tasks, data, information, scoring	Sec. 3
B	Implementation and system budgets	Secs. 4–5
C	History, frozen readouts, interventions	Sec. 5.4
D	Future targets, statistics, combinations	Secs. 5.2–5.3
E	Backbone and information flow	Secs. 4.4, 5.6
F	Pretraining, deployment, external tests	Sec. 5.6
G	Failures and qualitative evidence	Sec. 5.5
H	Training recipes and frozen decoding	Secs. 5.3, 5.5

A. Tasks, Data, and Evaluation

A.1. The Reported Fourteen-Task Suite

The completed studies use the migrated DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. The task-balanced score includes every task, including zero-success conditions.

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Later fresh-data production retains its own data identity and does not retroactively change the source of these results. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements (Table S1). The accepted FUTURE0, BASE2, and HISTORY1 packages reference the same archived evaluator source, task configurations, and fourteen-suite scenario manifest. We recover the definitions from that archive. Success is checked at each 2 ms physics step. Interrupted continuous conditions reset their counters. For T2, the sample completing the low-relative-motion counter also starts the stable-dwell counter; the two durations should not be added as an exact 0.600 s requirement.

Terminal events. The native result mapping groups events into five categories: *invalid* (invalid state); *boundary* (workspace or downstream exit, drop, expired goal window); *safety* (unsafe contact, excessive rim impulse, wire contact, stage block, release impulse, early robot contact, tray or gripper–support contact, striker trap); *task violation* (blocked driver, release outside the goal, path deviation or backtracking, premature release, stopping/deflecting without capture, unstable tipping, missing/extra impact, wrong striker contact, or insufficient outgoing motion change); and *timeout*. These categories describe evaluator outcomes.

There is no global priority among categories. Each family selects the first terminal event in its physics-step evaluation order and preserves that result. Grasp checks invalid state, workspace exit, downstream exit, and drop before success, then timeout. T1 inserts unsafe occluder contact after invalid state. T2 checks invalid state, rim impulse $> 0.10\,\mathrm{N\,s}$, drop, goal-window expiry, and release outside the goal before success and timeout. T3 checks invalid state, stage block, drop, workspace exit, wire contact, premature release, path deviation, and backtracking before success and timeout. Wire failure uses impulse $> 0.020\,\mathrm{N\,s}$ or penetration $> 0.004\,\mathrm{m}$ for more than two consecutive steps; path deviation is $> 0.030\,\mathrm{m}$ for 0.050 s, and backtracking exceeds 0.05 relative to maximum progress.

T4 checks invalid state, release impulse $> 0.002\,\mathrm{N\,s}$, unsafe ramp contact, unsupported tray contact, can tipping, early-contact/boundary branches, and invalid supported capture before success and timeout. A target that has exited the ramp, remains in contact with the downstream support, and has no fingerpad contact at speed $\leq 0.03\,\mathrm{m/s}$ for 0.25 s fails as stopping/deflecting only. T5 checks invalid state, unsafe contact, early robot contact, wrong striker, extra impact/striker trap, insufficient outgoing change, and boundary events before success. At the horizon, zero completed impacts yields missing selected impact; otherwise the result is timeout.

A.2. Scenario Sampling and Replay

The historical E7/agentview recipes specify proposal ranges (Table S2). Continuous min–max fields use uniform draws from NumPy PCG64; discrete choices are equiprobable before physical-feasibility checks. Evaluation loads each accepted scene's complete stored ScenarioSpec and does not resample its parameters. For harmonic translation,

$$p(t) = c + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0),$$

where $\phi \sim U[-\pi, \pi]$. Thus c differs from the initial position by the phase-dependent offset. T4 targets start at zero linear/angular velocity and evolve under free dynamics; no constant target speed is prescribed. Their slope is continuously sampled from $[5, 15]^\circ$. Ramp-local offsets use the ramp center as origin and lateral, downhill, and normal axes: $p_{\mathrm{world}} = c_{\mathrm{ramp}} + R_x(-\alpha)p_{\mathrm{offset}}$. The center is derived from the exit lip and ramp half-size; offsets are not added directly to the lip. T2-SE3 training uses $\omega_z \sim U[-.25, .25]$; validation excludes $(-.08, .08)$. The other thirteen tasks share train/validation proposal recipes.

Table S1. Native success conditions in the reported fourteen-task suite. Thresholds come from the evaluator source and task configurations bound to the accepted FUTURE0, BASE2, and HISTORY1 packages. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	Start with the native pre-grasp. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

Occlusion acceptance requires at least eight initially visible frames with visible fraction $\geq .95$, ten to sixty fully occluded frames (fraction $\leq .05$, lasting 0.5–3.0 s), and at least three reappearance frames with fraction $\geq .8$. These are acceptance criteria, not independent sampling variables. For example, the accepted OCC scenes have belt speeds 0.059588–0.099852 m/s within the wider proposal interval. Accepted source seeds can have gaps; a consecutive seed range does not reconstruct the scene set.

All 1,400 stored scenarios share the robot joint initialization

$$q_0 = [0,\ .19634954,\ 0,\ -2.61799388,\ 0,\ 2.94159265,\ .78539816]\ \text{rad},$$

and finger positions $(.020833, -.020833)$ m. T4 additionally stores a temporary preparation-target pose with $x, y \sim U[-.03, .03]$ and yaw $U[0, 2\pi]$, distinct from the release state.

The evaluator source stores recipes in `configs/scenarios/` and task bindings in `configs/tasks/`; the generation entry point is `se3_wam_benchmark/cli/generate_scenarios.py`. The evaluation payload contains the concrete `evaluation_reference/scenes/validation.jsonl` and fourteen `evaluation_reference/scenes/suites/sr_<task>.yaml` files. Each suite uses one repeat and filters the shared manifest by task. These files, with the stored scenario IDs and source seeds, define the replay set used for the paired comparisons.

The accompanying local material package at `results/replay_materials/` contains the archived evaluator, task and sampling configurations, all accepted scenarios, and both ordered selections. Portable copies change only

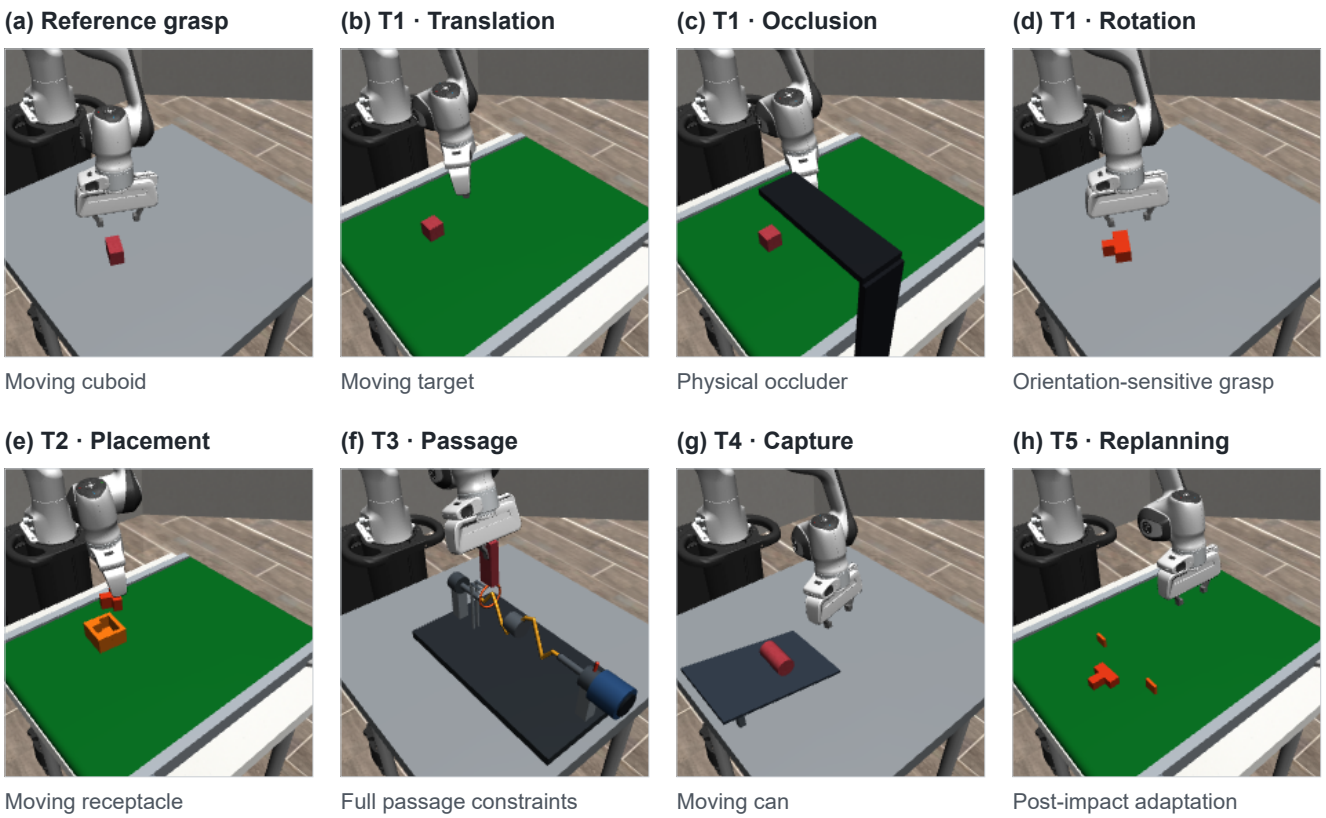


Figure S1. Recorded benchmark observations. Eight examples cover reference grasping; T1 translation, physical occlusion, and orientation-sensitive grasping; T2 moving-receptacle placement; T3 constrained passage; T4 capture; and T5 post-impact adaptation. For each displayed task, the lowest state index in the accepted DEV84 manifest supplies the final observed, uncropped 224×224 agentview RGB. Labels identify task conditions; the snapshots illustrate appearance and geometry, while task outcomes are reported separately. Table S1 lists all fourteen tasks.

execution paths; originals are retained. The archived manifest parser reads all 1,400 scenarios without changing their scientific fields, and the task/configuration loaders read all fourteen definitions. The supplied checker also verifies disjoint partitions, identities, seeds, and asset references. These checks parse the materials; physics and policy replay require the additional runtime and model assets listed in the package instructions.

A.3. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The combination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection indices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let b count candidate successes/reference failures and c candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b - c)/1260 \qquad \text{(S1)}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory.

We additionally compute 95% pointwise percentile intervals by resampling paired scenes within each task. For

Table S2. Historical validation proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

every replicate, each task retains ninety pairs (one hundred in the combination study), and the fourteen task differences are averaged with equal weights. Sampling a multinomial from each task's four paired outcome counts is equivalent to resampling its binary scene pairs. We use 100,000 replicates and the 2.5th/97.5th percentiles with linear interpolation. The generator is NumPy 2.3.5 PCG64; seeds are 20261017 plus the zero-based lexicographic comparison index for the twenty-seven future contrasts, and 20261017/20261018 for I versus F/E. The post-hoc I–E analysis excluding phase passage retains thirteen tasks with seed 20261019. Tasks are processed in lexicographic order. These intervals condition on the trained checkpoints and fixed task mixture; they do not include training-seed variation or candidate-selection uncertainty. They are not simultaneous intervals; the corrected McNemar family above remains the basis for the overall future-screen comparisons.

A.4. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. All one hundred T5 scenarios use a one-step command delay (50 ms); the other thirteen tasks use zero. The native action

queue returns a zero command until the delay elapses, then applies the oldest queued command. Forecast labels cover 50–400 ms relative to the source observation. Synchronous evaluation pauses the environment for policy computation, so it cannot measure the consequence of wall-clock observation age. Later fixed-delay and real-delay evaluations must retain their own time traces and report whether physics continues while the policy computes. Future labels follow the demonstration's recorded geometry under its executed actions. The current validity masks check geometric support, projection, and visibility. They do not establish that the same post-contact future remains valid for an alternative candidate action. Action-consistent label support is **[pending for online generation]**.

B. Reference Implementation and Budgets

B.1. Actual Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [26]. A 224×224 agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Projected image patches receive a view embedding, a linear embedding of mean valid metric depth, and a linear embedding of the patch-center robot-base XYZ before the view interface. All reported internal conditions use one agent view, so this interface returns that view directly and does not perform cross-view attention. XYZ is backprojected from fixed source pixels (indices 7, 23, …, 215) using camera intrinsics and extrinsics. These coordinates also enter 3-D RoPE for visual Q/K and the corresponding flow queries; object, robot, meta, and action tokens receive no XYZ. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The 25 object dimensions are $(p, \log R, \Delta p, \log \Delta R, \Delta t, v, \omega, a, \alpha)$, with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. The eight frames span -350 to 0 ms. Slots 0–6 encode adjacent intervals at their centers (-325 to -25 ms); only slot 7 retains the current absolute pose. Velocity divides interval displacement by elapsed time, and acceleration differences adjacent velocities at their interval centers. No state preceding the window is used. Each dimension is standardized by training statistics and invalid dimensions are zeroed. Three validity flags are appended before a $28 \to 768 \to 768$ MLP with GELU and a time embedding. Eight object, eight EEF, and 196 visual tokens form the scene bank. Sixteen learned meta tokens are separate inputs. The object-motion values come from simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens.

B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\text{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[\|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Training normalization uses the mean and population standard deviation of valid, non-repeated issued E7 commands in the training split, accumulated in FP64 and stored in FP32. Action flow matching uses a unit Gaussian in these coordinates. Each query samples $u \sim U[0, 1)$, shared across its full 8×7 action chunk. Deployment reverses normalization, decodes rotation vectors, and clips physical translation/rotation at 0.10 m/0.5 rad per axis; the gripper command uses its sign. Scene flow uses a metric Gaussian source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. AdamW uses learning rate 2×10^{-4}, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.05, and gradient-norm clipping at 1. After 1,000 warm-up updates, cosine decay spans the four-epoch schedule with a minimum learning-rate ratio of 0.1. Training uses FP32 with TF32 enabled, zero dropout, and an FFN expansion ratio of four. P03 weights

action, scene flow, and pose losses by one; Alpha is disabled. For cumulative scene flow F_h, the dynamic weight is $w_h = 1 + 128\,\mathrm{clip}(\|F_h - F_{h-1}\|_2/c, 0, 1)$, where $F_0 = 0$ and $c = 0.014620669186115265$ m. The weighted FM loss divides by three times the valid correspondence count, rather than by the sum of weights. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. All endpoints retain the declared exposure; topology transitions can change the subsequent numerical training trajectory.

B.3. Alternative Target Definitions

Table S3 specifies the target arrays after omitting batch and query axes. Unless stated otherwise, the eight horizons are 50, 100, ..., 400 ms. Surface correspondence follows fixed source geometry; a future occlusion does not by itself invalidate that identity. Track visibility is a separate label, using depth consistency within $\max(0.001\text{ m}, 0.001z)$ in the camera frame.

Wan receives nine RGB frames at 192×160 normalized to $[-1, 1]$. Its first temporal latent is dropped; the remaining two encode frames 1–4 and 5–8. Codec labels are extracted from actual decoder motion vectors. Target-track sampling starts nearest the mask centroid, then selects the farthest point; row-major order resolves ties. Fewer than 32 points are zero-padded and masked. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Regression objectives average squared error over channels and valid positions. Tracks add mean two-coordinate MSE and visibility binary cross-entropy with logits on valid correspondences. Valid positions are pooled across the batch, rather than averaging episode means. Native pose uses the geometric loss above. Replacement losses P04–P09, native flow, native pose, and action each have outer weight one.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision, each with weight 0.1. P15 combines the three losses at these same weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify every pairwise interaction. Raw total losses across these objectives are not commensurate.

B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [25]. DynamicVLA uses 50,000 updates at global batch 44, initialized from `hzxie/dynamic-vla-DOM` with vision, text, and connector frozen. It uses observations at indices $[-2, 0]$, a 20-step horizon, and normalization fitted on 800 demonstrations. OpenWAM uses a fresh fine-tuning run from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [23, 30]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [41]. DreamVLA receives seven causal RGB frames, EEF position/Euler orientation, two finger positions, and fixed task text. Its missing wrist view is black; inference uses DDIM10 and classifier-free guidance 1.5. OpenWAM receives current RGB, seven joint values, measured gripper opening, and task text. One agent view and two black wrist views form a 384×320 composite; state is mapped to its native 80-dimensional slots. Ten synchronous joint inference steps produce nine video frames and 32 actions, with video decoding disabled.

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, DynamicWAM sixteen, and OpenWAM thirty-two before replanning. DreamVLA predicts three steps but replans every tick, aggregates the latest three forecasts with exponential weights $\exp(-0.01k)$ over their age order, and executes one command. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

Table S3. Future targets in the representation screen. Shape excludes batch/query axes. Displacements are cumulative from the current state except codec vectors, which retain the decoder's native correspondence. All targets come from the recorded demonstration.

Target	Shape	Support, units, and validity
Scene flow	$196 \times 8 \times 3$	Fixed 14×14 patch centers; base-frame meters; valid source depth/geometry and finite future displacement.
Target pose	8×6	Cumulative translation (m) and left-relative rotvec (rad); valid object pose and query.
P04: Wan	$48 \times 2 \times 12 \times 10$	Frozen Wan2.2 VAE posterior mean and native normalization; future groups end at 200/400 ms; valid query.
P05: dense 2-D	$8 \times 2 \times 224 \times 224$	Fixed source pixels; displacement divided by 224; valid source and finite future projection.
P06: codec	$8 \times 2 \times 64 \times 64$	Each future/current pair is encoded as P/I frames; native vectors divided by 64; actual valid vector regions.
P07: scene tracks	$196 \times 8 \times 3$	Same patch source points; two displacement channels divided by 224 and one visibility logit; valid correspondences.
P08: object DINO	8×1024	Object-mask-area-weighted patch pooling; channel layer normalization; positive mask area.
P09: object tracks	$32 \times 8 \times 3$	Deterministic farthest-point sampling in the current target mask; displacement and visibility as P07; padded points invalid.
A_α: magnitude	196×8	Cumulative flow norm (m); same correspondence mask as flow.
P12: depth	$8 \times 1 \times 224 \times 224$	Metric future depth; finite positive pixels.
P13: DINO	$8 \times 196 \times 1024$	Frozen DINOv3-L/16 patches, channel layer normalization; valid query/horizon/patch.
P14: dynamic RGB	$8 \times 3 \times 224 \times 224$	RGB divided by 255; visible surfaces moving more than 1 mm in world coordinates, including robot and other geometry.

Table S4. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

System	Updates	Batch	Success /140	Execution / initialization note
ACT	50,000	32	50	Scratch; temporal aggregation
Diffusion Policy	50,000	32	54	Scratch; execute 8 actions
$\pi_{0.5}$	30,000	256	89	`pi05_base`; execute 10
DynamicVLA	50,000	44	24	Official DOM checkpoint; native continuous pipeline
OpenWAM-Alpha	34,490	192	76	Official Alpha foundation; execute 32
DynamicWAM	40,000	128	72	Official full initialization; execute 16
DreamVLA: 20 epochs	48,220	512	112	Independent schedule; aggregate forecasts, execute 1
40 epochs	96,440	512	114	Independent longer schedule
Study P00	40,104	16	88	Fresh task network; frozen DINO-L; execute 1
Study P03	40,104	16	93	Same reference recipe as P00
Study I	40,104	16	96	Fresh CNN condition; full visual history; execute 1

C. History Encoding and Conditional Information

C.1. Completed Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. All eight current-pose conditions are complete: current image, Window, Codec-MV, Flow, Flow12D, PACT-Age, Wan, and PACT-Uniform. The fresh GT-history reference also has complete training, 1,400-scene evaluation, and probe reports. It retains current RGB and adds privileged object-motion history. PACT-Age distributes fifty-six history queries from older to newer frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [21] through per-frame query compression, rather than reproducing its complete model.

All windows contain eight samples at 20 Hz, spanning -350 to 0 ms. Current/Window/PACT add fixed 768-dimensional sinusoidal frame-time embeddings in seconds; time subtraction is performed in FP64 before conversion to

Table S5. History encoders in the current-pose study. All conditions preserve the current 196 DINO-L visual tokens, T8 EEF states, and current object pose. History tokens are width 768 and carry no XYZ. Counts below concern added visual/numeric history tokens, not allocated scene slots.

Condition	Sampled times (ms)	History encoding	Added tokens
Current	0	Past image slots are masked.	0
Window	$-350{:}50{:}0$	Seven past frames, 196 DINO patches each; no query compression.	1,372
Flow	$-350, -250, -150, -50, 0$	Four $2 \times 64 \times 64$ flows; each RGB+validity field yields 64 patches and 14 learned-query outputs.	$56 + 4$
Flow12D	Same as Flow	Same visual encoder; four numeric slots receive twelve motion statistics and their flags.	$56 + 4$
Codec-MV	Same as Flow	Four actual codec MV fields, encoded by the same RGB+validity backend; no numeric slots.	56
Wan	$-350{:}50{:}0$	Repeat current frame once; $48 \times 3 \times 14 \times 14$ VAE latent $\rightarrow$ 147 patches $\rightarrow$ 56 queries.	56
PACT-Uniform	$-350{:}50{:} -50$	Shared eight-query bank reads each of seven past frames.	56
PACT-Age	Same as Uniform	Shared query bank interpolated to $[2, 4, 6, 8, 10, 12, 14]$ per frame.	56

FP32. Window retains every past patch. PACT uses one shared eight-query bank; age allocation linearly interpolates that bank with aligned endpoints, rather than adding frame-specific query parameters. Layer-normalized multi-head attention produces query-residual outputs.

Flow is bidirectional Farnebäck flow [10] on area-resized 64×64 RGB (pyramid scale 0.5, three levels, window 15, three iterations, polynomial neighborhood 5, sigma 1.2, flags 0). The four intervals last 100, 100, 100, and 50 ms. Validity requires finite vectors, in-frame endpoints, valid bilinear reverse sampling, and

$$\|f + b(x+f)\|^2 \leq 0.01(\|f\|^2 + \|b(x+f)\|^2) + 0.5. \quad \text{(S3)}$$

Flow color uses direction as hue, unit saturation, and magnitude divided by the interval's 99th percentile, clipped to $[0, 1]$, as value. Invalid pixels are black with validity zero. The four-channel field enters a $4 \rightarrow 768$ convolution with kernel/stride 8, followed by layer normalization, 64 learned spatial embeddings, and fourteen shared queries with interval-time embeddings. Attention, residual addition, and layer normalization produce fourteen tokens per interval.

For valid flow $f = (d_x, d_y)$, let $r = \|f\|_2$, bars denote pixel means, and q be the 99th percentile of r. Flow12D supplies

$$(\bar{d}_x, \bar{d}_y, \bar{r}, q, \Delta t, \bar{d}_x/\Delta t, \bar{d}_y/\Delta t, \bar{r}/\Delta t, q/\Delta t, a_x, a_y, \|a\|_2).$$

Acceleration differences adjacent mean velocities over interval-center time gaps of 100, 100, and 75 ms; the first interval has no acceleration. All dimensions except Δt are divided by 64, with no fitted mean/standard deviation. Twelve feature-valid flags form a 24-dimensional input to a $24 \rightarrow 768 \rightarrow 768$ GELU MLP. A separate $2 \rightarrow 768 \rightarrow 768$ MLP embeds the left/right times relative to the current frame. Pure Flow zeros the numeric values and flags but retains these four slots and their time embeddings. Thus the comparison adds magnitude, velocity, and acceleration statistics to a color encoding that normalizes magnitude.

Codec labels come from independent MPEG-4 Part 2 I/P pairs with no B frames, GOP 2, quantizer 2, YUV420p, one thread, and bitrate 10^6. Negated decoder backward vectors remain attached to destination blocks, unlike source-grid optical flow. Intra, uncovered, and out-of-frame-reference regions are invalid; no forward/backward confidence filter is applied. Wan uses a frozen FP32 Wan2.2 VAE posterior mean after right-padding T8 with the current frame. Its temporal groups end at $-350, -150, 0$ ms, with real-frame fractions $1, 1, 0.75$. A $48 \rightarrow 768$ kernel/stride-2 convolution creates 147 source tokens; learned spatial embeddings and an MLP for group time/fraction precede 56-query compression. VAE caches are cleared per window.

The original implementation allocates 1,584 scene slots for Current/Window, 272 for Flow/Flow12D, and 268 for the other compressed conditions. These counts include masked past-object slots; sixteen meta tokens are additional. Allocated slots and valid information are reported separately.

Each of the nine conditions has probe extraction over 350 episodes and 1,050 windows with 306 valid readouts. A completed CPU audit checks 9,450 saved NPZ files through 385,350 array comparisons: shared recorded inputs, labels, initial flow-matching noise, and the frozen current-image, geometry, robot-state, and clock baseline match across conditions. Oracle inputs match across the eight current-pose conditions; the intended GT-motion difference is excluded from equality checks. The completed point estimates give no visual-history gain over the current-image control. The nine full result files match on task, scene, source seed, and inference seed. Against Current, gain/loss counts are 137/223 for Window, 133/148 for Codec-MV, 176/204 for Flow, 183/208 for Flow12D, 132/246 for PACT-Age, 194/232 for Wan,

Table S6. Complete history results. The total is a development summary, not a separate held-out policy test.

Condition	/140	/1,260	/1,400	SR (%)
Current image	80	668	748	53.43
Window	69	593	662	47.29
Codec-MV	77	656	733	52.36
Flow	76	644	720	51.43
Flow12D	75	648	723	51.64
PACT-Age	65	569	634	45.29
Wan	70	640	710	50.71
PACT-Uniform	68	602	670	47.86
Reference with privileged target history				
Fresh GT history	84	740	824	58.86

122/200 for PACT-Uniform, and 238/162 for GT history. The main-paper intervals use Appendix A's task-stratified bootstrap with 100 paired scenes per task and 100,000 replicates. Seeds are 20261017–20261024 in this order. They describe scene uncertainty for the E4 seed-0 endpoints on the development pool, without training-seed, selection, or multiplicity adjustment. Outcome pairing and the separate probe-array audit cover distinct experimental records.

GT history has 586 joint successes and 414 joint failures with Current; its net gain is 76 scenes, or 5.43 points (95% interval 3.07–7.79). Table S7 shows a pronounced task tradeoff: XYZ grasping gains forty successes, while phase passage falls from 84/100 to 1/100. The aggregate score therefore does not imply a uniform gain from motion input.

C.2. Recipe Differences and Cross-Round Context

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at 2×10^{-5}. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect. Flow12D and Wan train fresh on sixteen PRO6000 devices; PACT-Uniform trains on eight PRO6000 devices, with a same-topology restart from update 781. Window, Codec-MV, and Flow resume sixteen-device A100 prefixes on PRO6000; PACT-Age additionally changes from sixteen to eight devices. All preserve global batch 16 and the four-epoch exposure. The device histories preclude a matched throughput comparison. GT history runs on eight PRO6000 devices until update 18,060, then resumes on sixteen with a native RNG reset. Its completed endpoint has 40,104 updates and 641,664 sampled chunks. This topology change is included in the interpretation of the single-training-seed comparison.

An earlier eight-condition study retained GT motion in every visual encoder (Table S8). Its current-image endpoint scores 835/1,400 (88/140 plus 747/1,260), distinct from the fresh 824/1,400 reference. No visual-history encoder exceeds the corresponding current-image point estimate in either round. Removing GT motion therefore does not reveal an overall visual-history advantage in the tested recipes. Cross-round differences remain descriptive: device, topology, and recovery histories differ, and each condition has one training seed. They do not identify a matched factorial interaction. Repeated current-pose/GT-motion × current-image/history training would assess its stability and interaction under a common topology.

C.3. Offline History Perturbations

For each of fourteen tasks, the probe protocol assigns 15/5/5 distinct episodes to training/validation/test, with three windows per episode. Table S9 uses the resulting 70 test episodes and 210 windows, with initial-noise seed 17. This is a probe split within the 350 policy-validation episodes, not a new policy test set. The intervention repeats the current image in all seven past slots or reverses the seven past images. Depth and calibration follow the same image permutation; the current frame, object information, robot history, timestamps, and masks remain fixed. These perturbed histories need not be physically realizable.

Each intervention re-encodes the window without a previous perception cache, rebuilds the condition, and reuses the actual Action/Flow initial-noise arrays. The native reports accept 840 same-input/noise controls per condition. Repeated-baseline and identity controls have zero differences; both current-image references have zero action and readout changes under every history intervention. Separate three-step checks in the eight current-pose conditions confirm exact condition-cache reconstruction and zero influence of masked old GT states. The GT-history reference intentionally consumes those states and is excluded from that masking check. Future labels are read only after prediction.

All seven history encodings change the issued first action under these perturbations, despite their lack of a success-rate gain in the main screen. For example, Window translation changes by 9.83 mm under repetition and 1.92 mm under reversal; Wan changes by 20.05 and 24.25 mm. The comparison distinguishes sensitivity to history from task benefit. It does not measure a closed-loop intervention or assign a mechanism to the success-rate differences.

This analysis aggregates the original per-task, episode-averaged reports. Their sample identities and source reports match the native records; the separate array audit above verifies the shared inputs and initial noise.

C.4. Frozen Readouts and Conditional Residuals

The readouts test which recorded motion variables can be decoded from frozen features. They use the same 15/5/5 episode split above, selected by sorted native episode identity.

Table S7. History-study successes, one hundred scenes per task and condition. The first eight conditions retain current GT target pose; GT motion retains current RGB and adds privileged target-motion history. All use one training seed and the fourth-epoch endpoint; the 1,400 scenes are the development pool.

Task	Current	Window	Codec	Flow	Flow12D	PACT-Age	Wan	Uniform	GT motion
Grasp	66	71	62	65	66	62	62	52	88
Belt	87	48	89	99	94	65	88	61	95
Occlusion	72	67	62	74	70	59	87	70	94
SO(3)	42	36	34	38	44	29	37	17	56
XYZ	41	48	54	53	47	55	62	44	81
SE(3) placement	0	0	0	0	0	0	0	0	0
Trans. placement	13	6	15	13	14	12	40	16	10
Full passage	0	0	0	0	0	0	0	1	0
Phase passage	84	6	99	0	0	28	3	67	1
Can	70	96	82	79	92	79	53	68	100
Slider	84	80	70	80	85	59	73	70	78
Sphere	41	55	46	65	76	38	63	45	66
Tabletop sphere	96	97	98	100	100	77	100	96	99
Replan	52	52	22	54	35	71	42	63	56
Total / 1,400	748	662	733	720	723	634	710	670	824

Table S8. Cross-round history context: successes over the same 1,400 scenes. The earlier round gives every encoder GT motion; the current-pose round removes it. Both retain current GT pose and robot history. Training-device and recovery histories differ, so this is not a matched factorial comparison. The earlier current-image endpoint (835) differs from the freshly trained GT-history reference (824).

Visual encoding	Earlier: GT motion	Current pose only
Current image	835	748
Window	834	662
Codec-MV	797	733
Flow	806	720
Flow12D	789	723
PACT-Age	816	634
Wan	782	710
PACT-Uniform	783	670

Table S9. Offline sensitivity in the eight current-pose conditions. Mean change in the issued first action, relative to the unchanged input, over 70 probe-test episodes (210 windows, fourteen tasks). Values are equal-task means of episode means. Translation is L2 change in mm; rotation is geodesic change in degrees. These are action changes, not action errors or success gains.

	Repeat current		Reverse past	
Condition	mm	deg	mm	deg
Current image	0.00	0.00	0.00	0.00
Window	9.83	1.28	1.92	0.34
Codec-MV	10.89	1.59	13.13	2.00
Flow	17.80	2.20	15.15	2.51
Flow12D	16.21	2.43	16.85	2.95
PACT-Age	6.07	0.48	2.17	0.25
Wan	20.05	2.70	24.25	3.98
PACT-Uniform	7.50	0.67	0.33	0.04

Ridge regression is fitted independently for each task, feature site, and label. The penalty is chosen from $\{0.1, 1, 10, 100\}$ by validation coordinate MSE, without refitting on validation or test data. Features use training-only means and population standard deviations, with values at most 10^{-8} replaced by one; targets are centered but not scaled, and the intercept is unpenalized. A shuffled control permutes the forty-five training-window labels with seed 1729, leaving validation and test labels unchanged.

Features and targets. The history site captures the visual-history encoder output entering the first Transformer layer: Window pools spatially within each past frame, while compressed encoders retain ordered tokens. The other sites are final Meta V and the input immediately before the action head at the first Action-branch function evaluation, $\tau = 0$, each mean/max pooled to 1,536 dimensions. The 18,048-dimensional conditioned baseline contains frozen current DINO-L features pooled to 4×4, current depth and validity, robot-base patch coordinates and validity, eight EEF states, relative times and masks, and the condition's actual oracle values and validity. Thus GT history supplies additional motion information despite equal baseline width. Past-position labels refer to the target body origin in the robot-base frame

at lags of 50, 200, and 350 ms. Velocity uses the last 50 ms backward difference; the future label is the demonstrated 400 ms continuation displacement.

Readout comparisons. A retained readout uses site features alone; a joint readout concatenates them with the conditioned baseline. Joint gain is baseline physical error minus joint physical error. For conditional residuals, the baseline penalty is selected on validation data, then five episode-disjoint folds produce training residuals: twelve episodes fit each baseline and three provide its held-out targets. Site features predict these residuals. Validation and test residuals instead use the baseline fitted on all fifteen training episodes. Reconstruction adds this fixed baseline prediction to the predicted residual. This tests a restricted decoder, not all information available to the policy.

Aggregation and uncertainty. Position error is the per-window Euclidean distance, averaged over three windows per episode, five test episodes per task, and fourteen equally weighted tasks. It is not RMSE or the model's pose-head error. Translation and velocity metrics cover all fourteen tasks; rotation and angular velocity cover only the three orientation-observable tasks. Per-task intervals resample the five episode means 2,000 times with seed 1729; they condition on the fitted probes and policy endpoints. Macro results average task point estimates and have no confidence intervals.

Results. Window has lower 350 ms position-error point estimates than Current at both Meta V (8.46 versus 10.89 mm) and the action input (8.31 versus 10.99 mm), while its success rate is 6.14 points lower (Fig. S2). Readable past state and sensitivity to image history therefore coexist with worse task performance in this comparison. The conditioned residuals give a similarly limited result: across three sites, nine conditions, and three labels—350 ms past position, last-50 ms velocity, and 400 ms future displacement—74 of 75 defined task-macro R^2 values are negative; the exception is Flow12D Meta V velocity at 0.034. Residual R^2 uses the test-residual mean as its reference, and its macro averages defined task scores rather than pooling tasks. A negative value does not establish absence of encoded information. Current pose is oracle-determined in every condition; GT history also determines past pose and velocity. Their unique-information residual scores are undefined, not zero, and baseline approximation errors are reported separately. The six GT-history past-position/velocity cells are therefore excluded from the count above. These readouts do not establish that the conditioned input is an information-theoretic sufficient statistic.

C.5. Frozen Object-Motion History Intervention

The completed follow-up fixes the fresh GT-history E4 endpoint and changes only its object-history input at inference. In the native object-descriptor path, every past target pose is replaced by the current pose before the native function recomputes interval kinematics. This *stationary history* removes past displacement, relative rotation, velocity, and acceleration before normalization, up to floating-point roundoff. Current target pose, RGB, robot history, timestamps, validity masks, and model weights are unchanged within each call. Later closed-loop observations follow each resulting trajectory; they are not held equal across interventions. The replacement follows the temporal frame cache, and conditioning is rebuilt so cached descriptors cannot restore the original motion.

Controls and sampling. The action test reuses the 70 test episodes and 210 windows of the probe split, with initial-noise seed 17 and the same actual noise arrays for native, identity-copy, and stationary-history calls. Identity copying preserves all 210 action arrays elementwise. Three consecutive online windows from one episode additionally verify full recomputation, incremental temporal caching, and conditioning-cache reconstruction. In closed loop, identity copying reproduces all original fixed140 success/failure labels, including its 84 successes. The follow-up then evaluates stationary history on all 1,400 original scenes, preserving scene identities, source/inference seeds, five Euler steps, and the original execution interface. The true-history 1,400-scene result is reused from the accepted endpoint; no training or additional true-history evaluation is counted.

Action response and closed-loop dependence. All 210 first issued actions change under stationary history. Mean first-translation change is 18.40 mm (95% interval 16.90–19.93), and mean rotational change is 0.0424 rad (0.0368–0.0482). Success falls from 824/1,400 (58.86%) to 9/1,400 (0.64%): 815 originally successful scenes fail and no originally failed scene succeeds. The paired drop is 58.21 points (95% interval 56.50–59.86). The remaining successes are seven reference-grasp scenes and two orientation-sensitive-grasp scenes. Rigid-motion placement and full passage remain unsuccessful in both conditions. Figure S3 shows the action distribution and all fourteen tasks.

The follow-up retains its native 20,000-replicate bootstrap with seed 1729, separately from the main history table's 100,000-replicate analysis. Closed-loop intervals resample paired scenes within each of fourteen fixed tasks. Action metrics first average the three windows of each episode, then resample the five test episodes within each task and average tasks equally. Thus the action interval has seventy episode units, not 210 independent windows. Both intervals

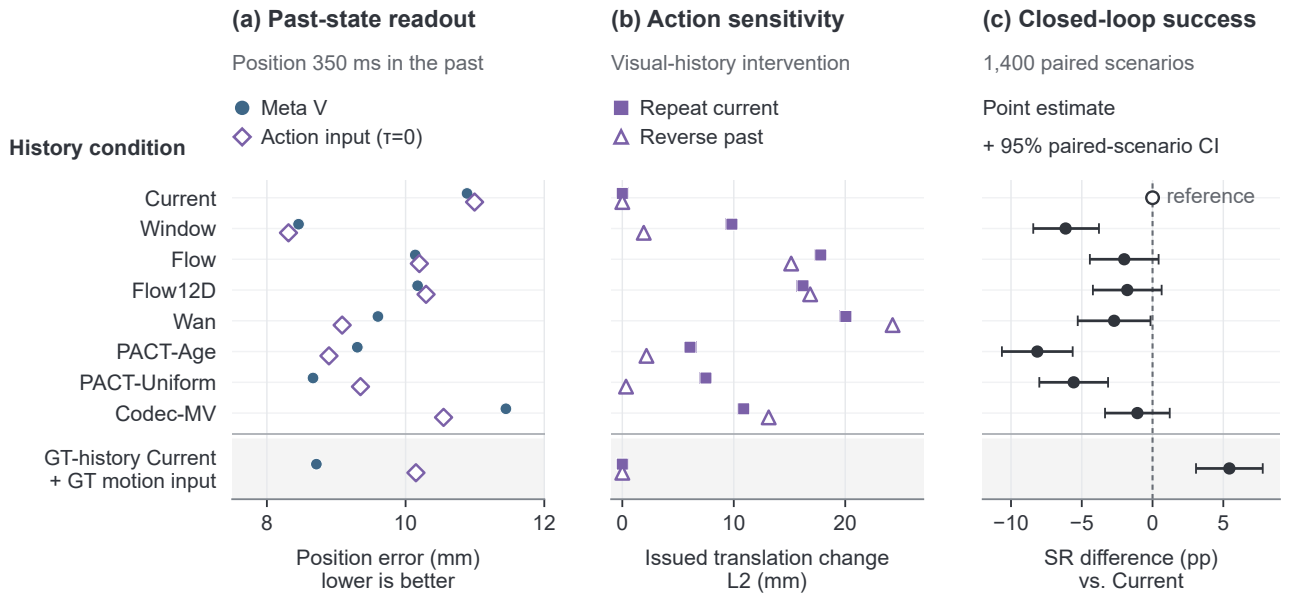


Figure S2. History decodability, action sensitivity, and closed-loop success. (a) Frozen ridge error for position 350 ms in the past, from final Meta V or the first action-head input ($\tau = 0$). (b) L2 change of the first issued translation after action limits, under repeat-current or reverse-past intervention with identical initial noise. Panels (a,b) show descriptive means across fourteen fixed tasks, without confidence intervals. (c) Success-rate difference from Current over 1,400 paired scenarios, with pointwise 95% intervals from 100,000 task-stratified resamples. These intervals condition on single-seed checkpoints and the development pool. The separated GT-history reference receives additional GT motion. Both current-image references have no visual history to perturb; GT motion remains intact. Lower readout error and nonzero action sensitivity do not establish a task benefit.

condition on one frozen endpoint and exclude training-seed uncertainty.

These controls establish that this endpoint uses motion history in action generation and closed-loop control, and that the learned policy is sensitive to its removal at inference. The 58.21-point loss measures this input mismatch; Current trained without GT motion still achieves 53.43%. Together with the training comparison and visual-history probes, this supports retaining structured motion history in the reference recipe. It does not establish SE(3) sufficiency, mediation of all visual history through SE(3), or a compute advantage. Translation/rotation contributions and robustness to estimated, delayed motion histories remain separate questions.

C.6. Further Matched-Current and Occlusion Diagnostics

The stationary-history intervention tests one specific input mismatch. Valid alternative histories require matching the current target pose, robot state, view, and task condition while changing valid preceding motion. Matching tolerances, group counts, and history-necessary subset acceptance remain to be defined for these **[pending]** follow-ups. A substitute that changes the current state tests an additional intervention. Occlusion tests must report visibility and duration in physical time, while sampling-gap tests preserve the meaning of displacement by retaining elapsed time. The planned outputs link motion decoding, action change, contact outcome, and unconditional task success. No-contact failures remain in the success denominator rather than receiving zero contact error.

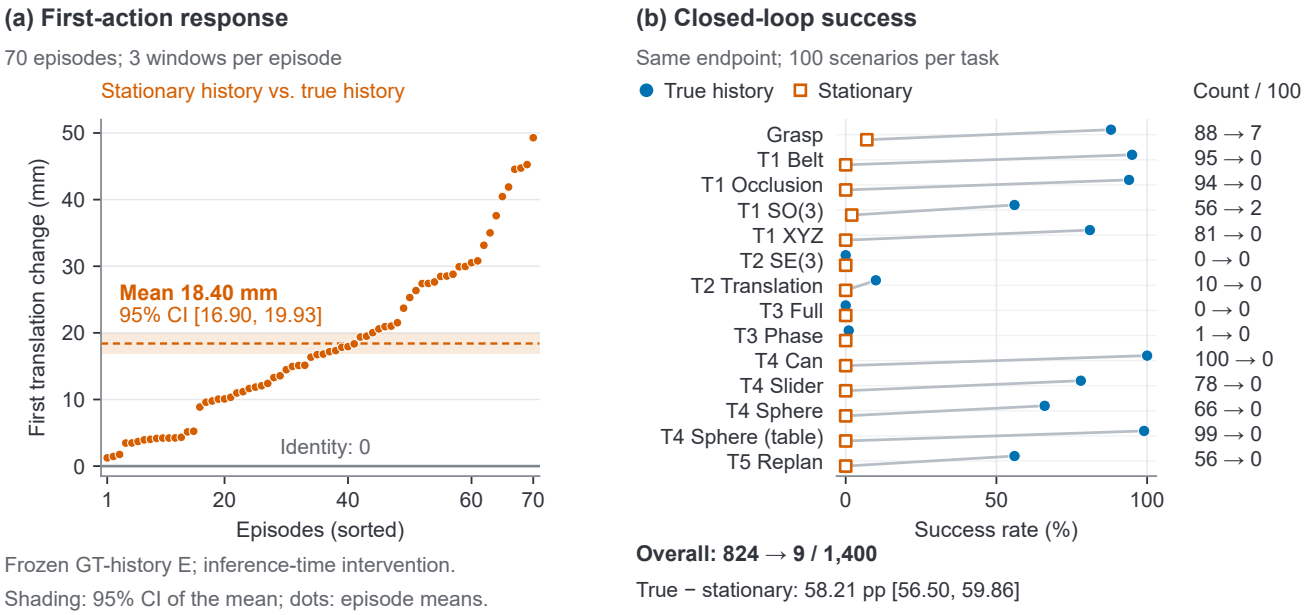


Figure S3. A frozen policy depends on correct object-motion history. Stationary history repeats the current target pose in past slots and recomputes kinematics; other input paths are retained within each call. (a) L2 changes in the first issued translation after action limits. Each point averages three probe windows from one of seventy episodes; the mean and 95% interval use task-stratified episode resampling. Identity copying changes no action. (b) True versus stationary history on fourteen tasks with 100 paired scenes each. The aggregate paired drop uses task-stratified scene resampling. Both intervals use 20,000 replicates, seed 1729. True-history outcomes are reused from the original endpoint; identity copying separately reproduces its fixed140 success/failure labels. This inference-time mismatch effect is distinct from the 5.43-point training-comparison gain of adding GT motion.

D. Future Supervision and Combination Selection

D.1. All Fifteen Future Conditions

All conditions in Table S10 have action supervision. Scene-flow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates with task-stratified paired bootstrap intervals, as specified in Appendix A.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

D.2. Paired Outcomes and Task Heterogeneity

Table S11 gives gains, losses, difference intervals, and adjusted exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S12 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy failure remains in the 1,260-scene denominator.

D.3. Combination Selection

The next screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine sched-

Table S10. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

		Development	Disjoint complement	
ID	Future supervision	Success / 140	Success / 1,260	SR (%)
Reference and component objectives				
P00	No future supervision	88	791	62.78
P01	3-D scene flow	95	782	62.06
P02	Target pose	92	789	62.62
P03	Scene flow + target pose	93	786	62.38
Scene replacements: target-pose objective retained				
P04	Wan latent + pose	92	783	62.14
P05	Dense 2-D displacement + pose	87	746	59.21
P06	Codec motion + pose	86	795	63.10
P07	Sparse scene tracks + pose	90	807	64.05
Target replacements: scene-flow objective retained				
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target tracks	88	808	64.13
Additional objectives: scene flow + target pose retained				
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65

Table S11. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Intervals are pointwise task-stratified paired bootstrap intervals for the SR difference, conditional on the trained checkpoints (Appendix A). Exact two-sided McNemar tests use the full family of fourteen P00 and thirteen P03 contrasts: $p_{\mathrm{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

	P00: no future supervision				P03: scene flow + target pose			
ID	Gain	Loss	95% CI (pp)	Adjusted p	Gain	Loss	95% CI (pp)	Adjusted p
Reference and component objectives								
P01	80	89	$[-2.70, 1.27]$	1	72	76	$[-2.14, 1.51]$	1
P02	61	63	$[-1.83, 1.51]$	1	61	58	$[-1.43, 1.90]$	1
P03	64	69	$[-2.14, 1.35]$	1	—	—	—	—
Scene replacements: target-pose objective retained								
P04	74	82	$[-2.54, 1.27]$	1	71	74	$[-2.06, 1.59]$	1
P05	53	98	$[-5.40, -1.75]$	0.00844	47	87	$[-4.92, -1.43]$	0.0188
P06	81	77	$[-1.59, 2.22]$	1	68	59	$[-1.03, 2.46]$	1
P07	85	69	$[-0.63, 3.17]$	1	81	60	$[-0.16, 3.49]$	1
Target replacements: scene-flow objective retained								
P08	75	72	$[-1.59, 2.06]$	1	59	51	$[-0.95, 2.22]$	1
P09	91	74	$[-0.56, 3.25]$	1	79	57	$[0.00, 3.49]$	1
Additional objectives: scene flow + target pose retained								
A_α	60	73	$[-2.78, 0.71]$	1	53	61	$[-2.30, 1.03]$	1
P12	73	73	$[-1.83, 1.83]$	1	63	58	$[-1.27, 2.06]$	1
P13	70	73	$[-2.06, 1.59]$	1	60	58	$[-1.51, 1.83]$	1
P14	71	72	$[-1.90, 1.75]$	1	62	58	$[-1.35, 2.06]$	1
P15	74	63	$[-0.95, 2.70]$	1	63	47	$[-0.32, 2.94]$	1

Table S12. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

	Reference	T1: target motion				T2: placement		T3: passage		T4: capture				T5	
ID	Grasp	Belt	Occl.	SO(3)	XYZ	SE(3)	Trans.	Full	Phase	Can	Slider	Sphere	Tabletop	Replan	Total
Reference and component objectives															
P00	85	90	90	75	76	0	8	3	15	90	69	45	86	59	791
P01	84	90	90	64	69	0	18	0	11	90	74	46	90	56	782
P02	87	90	90	78	73	0	7	0	0	90	71	59	89	55	789
P03	86	90	88	75	77	0	7	0	6	86	80	50	87	54	786
Scene replacements: target-pose objective retained															
P04	83	85	84	73	76	0	20	0	0	90	72	54	89	57	783
P05	84	89	90	66	75	0	10	0	0	90	71	23	88	60	746
P06	86	88	89	79	70	0	6	4	2	89	72	60	85	65	795
P07	82	90	90	78	71	0	17	0	0	89	79	55	90	66	807
Target replacements: scene-flow objective retained															
P08	84	90	90	71	81	0	13	0	0	90	77	60	90	48	794
P09	84	90	89	65	78	0	21	0	0	90	77	59	90	65	808
Additional objectives: scene flow + target pose retained															
A_α	86	90	90	72	75	0	3	0	0	85	73	55	90	59	778
P12	83	90	90	76	76	0	5	0	0	89	77	57	90	58	791
P13	86	90	90	80	73	0	6	0	0	88	78	54	90	53	788
P14	85	89	90	72	79	0	5	0	0	90	80	53	88	59	790
P15	87	90	90	79	81	0	11	0	0	90	75	54	90	55	802

Table S13. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

ID	Future supervision	Boundary	Safety	Task viol.	Timeout	Total
Reference and component objectives						
P00	No future supervision	84	297	17	71	469
P01	3-D scene flow	62	324	23	69	478
P02	Target pose	77	331	17	46	471
P03	Scene flow + target pose	76	295	51	52	474
Scene replacements: target-pose objective retained						
P04	Wan latent + pose	79	310	41	47	477
P05	Dense 2-D displacement + pose	87	356	14	57	514
P06	Codec motion + pose	96	250	23	96	465
P07	Sparse scene tracks + pose	47	329	20	57	453
Target replacements: scene-flow objective retained						
P08	Scene flow + object DINO	61	322	42	41	466
P09	Scene flow + target tracks	67	299	35	51	452
Additional objectives: scene flow + target pose retained						
A_α	P03 + motion magnitude	57	315	65	45	482
P12	P03 + depth	62	327	40	40	469
P13	P03 + DINO	71	326	27	48	472
P14	P03 + dynamic RGB	70	322	18	60	470
P15	P03 + depth / DINO / dynamic RGB	59	343	19	37	458

ule. Its twelve candidates and the selected configuration's matched P00 and P03 controls are complete. All fourteen conditions start with sixteen devices and global batch 16. L and N resume on eight devices with batch 2 per device, preserving the optimizer, schedule, normalization, sample cursor, and 40,104-update budget, but resetting destination-rank random states. K resumes on a different GPU type with sixteen devices. M completes training on sixteen devices and moves only its remaining 1,260-scene evaluation to eight. I, E, and F retain sixteen-device training. The full grid thus matches nominal exposure, but its P09 candidates do not share one hardware and random-state history.

CNN and temporal conditioning. The trainable CNN has four convolution–GroupNorm–GELU blocks with channels $3 \to 32 \to 64 \to 128 \to 192$. Each convolution has kernel 4, stride 2, padding 1, and no bias; GroupNorm uses eight groups. The 224×224 image becomes 14×14 patches, followed by token LayerNorm ($\epsilon = 10^{-6}$) and a $192 \to 768$ projection. RGB is divided by 255, bilinearly resized with antialiasing, and normalized by means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Geometry embeddings follow Appendix B. The full T8 window retains $8 \times 196 = 1,568$ visual tokens, eight EEF tokens, and eight object-kinematic tokens: 1,584 scene tokens when the full window is available, plus sixteen learned meta tokens. Relative-time embeddings and trunk attention fuse the window; there is no history-query compression or temporal mean pooling.

Current-target reconstruction. The mean of the final sixteen meta tokens enters a $768 \to 256 \to 3,072$ MLP with GELU and final sigmoid, producing $3 \times 32 \times 32$ RGB. Simulator target geometry IDs select the currently visible target and its tight bounding box. The crop is resized to 32×32 (bilinear RGB, nearest-neighbor mask); values outside the mask are zero. Loss is masked L1 with outer weight 0.1, divided by the global count of valid RGB elements. An empty mask contributes zero without dropping the training sample. The reconstruction is a training readout, not an action input. I, E, and F share this path, the same D backbone, and the same training normalization from 1,324,374 valid issued actions in 11,200 training episodes. The 350 validation episodes do not contribute normalization statistics.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S14 shows. The selected candidate is carried forward for testing across training seeds.

The matched P00 control F retains I's trainable CNN, full visual window, privileged motion input, and current-target masked RGB reconstruction (weight 0.1), with future objectives disabled. Both use fresh seed 0, global batch 16, four epochs, and the full-run cosine schedule. F obtains 79/140 and 731/1,260 successes, totaling 810/1,400. For I versus F, the paired counts are 746 joint successes, 189 I-only successes, 64 F-only successes, and 401 joint failures. The 125-scene net gain is 8.93 percentage points (95% pointwise interval 7.00–10.86). It includes 72 on phase passage and 22 on replanning; translational placement loses seven. These are results on the pool used to select I; the interval does not correct for candidate selection.

The matched P03 control E changes the future objectives to scene flow and target pose. It obtains 87/140 and 774/1,260 successes, totaling 861/1,400. I versus E has 786 joint successes, 149 I-only successes, 75 E-only successes, and 390 joint failures. The net 74-scene gain is 5.29 percentage points (95% pointwise interval 3.50–7.07). Phase passage contributes 72 of these net successes; the remaining thirteen tasks contribute two in total. The aggregate difference is therefore highly task-concentrated within this development pool. A post-hoc calculation excluding phase passage gives 863/1,300 for I and 861/1,300 for E: 0.15 points (interval $[-1.69, 2.00]$). This descriptive sensitivity analysis does not replace the predefined fourteen-task score. Confirmation requires unused scenes and additional training seeds.

D.4. Granularity and Prediction-to-Action Evidence

A future comparison must separate role coverage, variable choice, spatial allocation, and compute. The fifteen-target screen changes target packages, not only allocation at equal budget. A direct fine-target/coarse-context test still requires a compatible uniform reference with the same identity information, horizon support, and declared token/compute budget. Its result is **[pending controlled granularity comparison]**.

Physical readouts should compare target translation and rotation at each horizon; flow and tracks require valid correspondences and visibility. Their association with contact error is a separate analysis. The accepted source contains readout arrays, but this manuscript does not substitute flow-matching validation loss for freely generated physical error. The aligned prediction/contact figure requires **[generated predictions, actual states, and contact events]**. If a future oracle is introduced, it must use the same action interface and target times. Any oracle-gap ratio is reported only with a supported, non-negligible denominator.

Table S14. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. E and F are the selected configuration's matched scene-flow/pose and no-future controls. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

	P15: auxiliary				P07: scene tracks				P09: target tracks				P03	P00
Task	DINO − (B)	DINO + (A)	CNN − (D)	CNN + (C)	DINO − (H)	DINO + (G)	CNN − (J)	CNN + (I)	DINO − (L)	DINO + (K)	CNN − (N)	CNN + (M)	CNN + (E)	CNN + (F)
Reference grasp														
Grasp	90	89	91	92	90	92	87	90	79	85	77	87	85	78
T1: target motion														
Belt	100	100	99	100	100	100	100	97	99	100	90	99	100	100
Occlusion	100	100	99	99	99	100	100	97	96	99	82	98	99	98
SO(3)	86	71	80	88	75	87	69	86	69	62	64	78	78	71
XYZ	58	58	82	75	73	75	89	82	44	56	82	84	78	82
T2: receptacle motion														
SE(3)	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Translation	8	4	6	5	9	1	14	4	4	5	9	14	9	11
T3: constrained passage														
Full	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Phase	0	0	0	0	0	0	28	72	0	0	0	2	0	0
T4: object capture														
Can	99	99	99	98	100	99	99	100	97	99	96	99	99	99
Slider	88	85	88	91	74	83	86	81	84	84	88	88	84	77
Sphere	71	67	67	63	47	40	53	56	68	68	62	64	68	45
Tabletop sphere	100	100	100	99	99	100	95	98	100	100	100	100	99	99
T5: post-impact replanning														
Replan	73	63	64	57	74	72	76	72	80	72	79	57	62	50
Total / 1,400	873	836	875	867	840	849	896	935	820	830	829	870	861	810

E. Backbone, Information Flow, and Cross-Checks

E.1. D/E Backbone Comparison

The planned backbone comparison uses the confirmed complete history/future configuration, retaining the reference G_0 program where action does not read predicted future. D shares QKV and output projections across WM, Flow, and Action, with a separate feed-forward network for each. Adapted E assigns QKV/output projections, adaptive layer normalization (AdaLN), and feed-forward networks to two parameter groups: World and Action. Scene, meta, and any active future-Flow tokens share World parameters; Action has its own group. World observation/meta tokens use diffusion time $\tau = 0$, while Flow keeps its per-token time. For a future target decoded without scene-Flow flow matching, both candidates omit the active Flow branch.

The comparison also retains each candidate's internal normalization: E's World AdaLN at $\tau = 0$ has randomly initialized scale/shift biases, whereas D's WM LayerNorm starts with unit affine scale and zero shift. Thus, the shared input normalization accompanies a comparison of parameter organization and internal normalization, rather than isolating feed-forward sharing.

The planned two-arm comparison trains fresh seeds 0, 1, and 2, with the same initialization sources, data exposure, optimizer schedule, and endpoint. For each seed, common input projections, history/time parameters, and readouts start identically; the backbone parameter groups are initialized separately. Both arms require fresh training under this paired initialization. If the current four-epoch recipe is retained, there are 240,624 updates across six runs and 8,400 evaluation episodes at 1,400 per endpoint. These are planned counts. The active-backbone parameter difference is targeted within 5%; frozen frontend parameters and their runtime costs are recorded separately. Matching exposure or parameter count does not match training FLOPs.

The predeclared practical SR and non-inferiority margins are two percentage points, with a 10% p95 latency benefit considered for cost selection. Intervals must support the corresponding superiority, equivalence, or non-inferiority criterion before that label is used. If three seeds are inconclu-

sive, both arms may be expanded to five under the existing plan. Same-device public-policy timing includes internal perception/action computation and excludes external rendering and oracle access. Selection and uncertainty are **[pending]**.

E.2. Seven Information-Flow Conditions

R0 versus R1 tests auxiliary supervision under action-only deployment. R1's two deployment modes test the cost and effect of generating an unused future from the same checkpoint. R1–R4 examine information access. R2 versus R5 and R3 versus R6 compare synchronous and sequential programs. Sequential second stages train on clean first-stage labels and deploy on first-stage predictions. The corresponding train/test mismatch belongs in the interpretation.

The current generative state is scene flow with shape $[B, S, 196, 8, 3]$; pose, tracks, depth, DINO, and RGB can be deterministic auxiliary readouts. A selected sparse-track or Wan representation requires its own generated-state, normalization, noise, and loss definition before this experiment can run. Replacing it with scene flow would change the question. This interface is **[pending final representation binding]**.

The planned three seeds yield twenty-one trained configurations and twenty-four deployment configurations, including R1's second mode; at 1,400 scenes each, that is 33,600 rollout episodes. Alpha is fixed to zero. Per-modality Euler5 is common, but joint inference uses five combined calls and sequential inference uses five plus five. Actual calls, latency, parameters, training exposure, and valid readouts are reported alongside SR.

E.3. Mechanisms and Final Design

An independently replaceable future interface permits oracle, valid matched-current swap, time-shift, or calibrated-noise diagnostics. Each intervention holds perception, robot state, and action-source noise fixed. Changes in action alone show sensitivity; evidence for useful forecast use also requires task and safety outcomes. Removing a branch at inference and retraining without its loss answer different questions.

After selecting information flow, the study revisits retained future representations and applies the selected flow to the unselected backbone. History choices are revisited when the future target changes. Only the necessary comparisons are repeated; previously valid matched endpoints can be reused. Ranking reversals change the conditional design choice. Multi-seed and scenes not used for selection are required before passing a complete (H, Z, B, G) design to pretraining. The selected configuration, cross-check estimates, and mechanism readouts remain **[pending]**.

Table S15. Planned Q3 conditions; no measured result is assigned to these rows. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

ID	Future loss	$F \to A$	$A \to F$	Cross-stream training input	Inference order
R0	No	—	—	None	A only
R1	Yes	—	—	None	A only
R1	Yes	—	—	None	Joint F and A
R2	Yes	Yes	—	Synchronous noisy states	Joint F and A
R3	Yes	—	Yes	Synchronous noisy states	Joint F and A
R4	Yes	Yes	Yes	Synchronous noisy states	Joint F and A
R5	Yes	Yes	—	Clean F for second-stage A	F, then A
R6	Yes	—	Yes	Clean A for second-stage F	A, then F

F. Pretraining, Cost, and External Evaluation

F.1. Measured Policy Cost and Physical Delay

The future-screen policy call means span 177.071–184.212 ms; p95 spans 178.726–185.133 ms; peak allocated memory spans 1.571–1.617 GiB. These are same-device, fixed-order measurements including internal perception and action computation. Loading, external oracle reads, I/O, rendering, and priming are excluded. Each arm uses three recorded eight-frame sequences, six warm-up calls, and thirty measured calls (ten per sequence). After each reset, seven real calls prime the observation history; the eighth is timed with CUDA synchronization before and after the call. The sequences cover translation, occlusion, and post-impact replanning. Inference is FP32 with TF32 enabled, batch/sequence dimensions $B = S = 1$, and five Euler action steps. NVML identifies the shared device as `NVIDIA RPBZZZ6`, with 102,641,958,912 bytes of memory; CPU affinity is 0–2. Memory is the post-warm-up PyTorch allocation, including resident weights. One measurement round supports this range; differences of a few milliseconds do not establish a robust speed ranking.

For deployed commands, action age is the difference between sensor capture and command application. The planned deployment profile separates capture, availability, preprocessing, inference, queuing, issue, and application. It records hardware, precision, batch size, warm-up, synchronization, and timed sample counts. Fixed-delay simulation tests a controlled lag; wall-clock execution tests the actual combined system. The final same-device profiles, deadline misses, and action-age distributions are **[pending]**.

F.2. Initialization Controls

Three predefined controls separate initialization from downstream design. For DreamVLA, the control removes the LIBERO policy checkpoint while retaining the MAE/CLIP initialization, teacher targets, losses, and the independent twenty-epoch recipe (48,220 updates, global batch 512). For $\pi_{0.5}$, the control replaces the base initialization with the LIBERO expert while retaining the 30,000-update, global-batch-256 downstream recipe. These comparisons use the original fourteen-task evaluation partitions.

A separate transfer study uses an action-only version of the four-layer, width-768 D trunk with a trainable CNN, T8 RGB and robot-state history. It removes target-pose input, depth/camera geometry, the flow stream, and all future and target-region reconstruction objectives. The two downstream conditions share this network and normalization fitted on the same 11,200 demonstrations; one starts fresh and the other loads LIBERO-90 visual-motor pretraining [20]. The pretraining protocol uses 4,050 demonstrations, 450 held-out demonstrations, thirty epochs, and 8,520 updates at global batch 256. Both downstream runs reset optimizer and scheduler state and use four epochs, 40,104 updates, and global batch 16. This contrast tests visual-motor transfer within the simplified policy; it is neither a P07 pretraining ablation nor a language-conditioned LIBERO evaluation. Endpoint success rates for these three controls are **[pending]**. They do not supply the formal robot-data base described below.

F.3. Recipe and Robot-Data Base

After the design cross-checks, the study fixes input modalities, history support, future targets/horizons, information flow, and action semantics. The next stage optimizes the existing loss weights and training recipe, and changes depth/width within the chosen architecture for scale comparison. Changing the representation or adding an information path belongs to a new design comparison rather than an optimizer ablation. Appendix H reports a completed recipe comparison on development combinations G and I; it does not replace the planned study of the final selected design.

DROID supplies real robot demonstrations [15]. Preparation checks sensor timing, calibration, depth, command versus measured motion, gripper semantics, and training-only normalization. It does not provide object pose GT; the target-pose pipeline must be separately validated for any selected

pose-based objective. Only targets used by the selected design and necessary ablations need to be produced. The data report includes valid-label masks and rejected-action rates. This stage is **[planned]**.

The formal base is trained on real robot data. Two independent post-training branches start from the same base: one for simulation and one for real-robot tasks. Simulation-from-scratch experiments supply recipe experience and a strong reference, while their weights do not initialize the formal robot-data base or the real-robot branch. Comparisons with and without pretraining share downstream data and exposure, and additionally allow equally budgeted recipe searches. They report the pretraining cost explicitly. Data, model, and compute scale have separate axes.

F.4. Planned External and Real-Robot Evidence

Single-arm DOMINO [9] tests dynamics under a different task/data protocol. Static compatibility uses an explicitly selected native suite such as LIBERO [20]. Real conveyor manipulation evaluates perception, prediction, action, and timing together. The final phase reuses completed pretraining, scaling, and independent adaptation results, then fills uncovered object, scene, motion, and delay conditions. It does not repeat an already completed scaling matrix.

The real-robot protocol must specify objects, speed/delay strata, trial allocation, ordering, calibration, safety termination, and human intervention. External native scores are reported separately from the fourteen-task simulation SR. Final training scale, robot trials, and real-world performance are **[pending]**.

G. Diagnostics and Qualitative Evidence

G.1. Supervision at the Executed Action Position

All 800 historical full-passage demonstrations release only at the final action: chunk position h_7 has release supervision, while the executed position h_0 has none. This identifies an action-position coverage gap.

Tail-window training produces 2/16 successes in diagnostic validation and 1/6 in a selected-scene follow-up. For phase passage, three seeds after 256 updates give 0/96, 12/96, and 26/96; at 1,024 updates all three give 0/96. Six first-action-weighting endpoints total 576 failures in 576 trials, and six GT-role diagnostic endpoints also total zero successes in 576. These local budgets are separate from the fourteen-task study.

A zero action-source deployment intervention changes the three 256-update scores from 0/12/26 to 0/1/46 successes out of 96 each. This post-hoc intervention is exploratory. A reproducibility investigation links some same-seed trajectory branching to one-gray-level RGB changes; fixed full-input/RNG inference and fixed-action replay remain consistent.

G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at ± 0.5. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection, and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. Neither is adopted as the production default in the evidence cited here.

The broader fourth-to-eighth-epoch dead-zone report has not passed strict historical trajectory-consistency checks and is excluded from main-table claims. Decoder effects and representation effects require separate comparisons. Appendix H reports the subsequent frozen-head study and its failed regression gate.

G.3. Offline Target-Pose Predictions

Figure S4 compares deterministic fourth-epoch P15 pose readouts with two kinematic baselines on 84 states (six per task). All eight horizons are valid, yielding 672 state–horizon pairs. Quantiles use linear interpolation with equal state weights; states from one episode may be dependent. Median translation error rises from 1.45 mm at 50 ms to 4.04 mm at 400 ms; the corresponding rotation medians are $0.47°$ and $1.07°$. The cases are selected by mean translation error, independently of task success.

Kinematic baselines. CV and CA use the final two and three observed GT poses, respectively, within the same history available to P15. They do not read future labels or instantaneous simulator velocities. The saved raw positions, xyzw quaternions, and FP64 timestamps verify a 50 ms interval d for all states. The resulting last-interval derivatives agree with the model's unstandardized object-input values. For positions p_{-2}, p_{-1}, p_0 and a forecast horizon δ in seconds, CV predicts $\delta(p_0 - p_{-1})/d$. CA predicts $\delta v_0 + \delta^2 a/2$, where

$$v_0 = \frac{3p_0 - 4p_{-1} + p_{-2}}{2d}, \qquad a = \frac{p_0 - 2p_{-1} + p_{-2}}{d^2}.$$

CV retains the final interval velocity; CA extrapolates it from the -25 ms interval center to the current time.

In the fixed robot-base frame, define $\omega_+ = \mathrm{Log}(R_0 R_{-1}^\top)^\vee/d$ and $\omega_- = \mathrm{Log}(R_{-1} R_{-2}^\top)^\vee/d$. Rotational CV uses $\Delta R(\delta) = \mathrm{Exp}(\delta\omega_+^\wedge)$. CA sets $\omega_0 = 1.5\omega_+ - 0.5\omega_-$ and $\alpha = (\omega_+ - \omega_-)/d$, and integrates to $t = \delta$:

$$\dot{\Delta R}(t) = [\omega_0 + t\alpha]_\times \Delta R(t), \qquad \Delta R(0) = I.$$

Exponential midpoint steps are at most 1 ms; halving the step changes any predicted rotation by at most $0.000548°$. There is no smoothing, clipping, or fitted baseline parameter.

Table S16. DEV84 offline prediction errors: 84 states and eight horizons, yielding 672 dependent observations. CV/CA use the last two/three GT poses; P15 also uses the full history and its other observation inputs.

	Translation (mm)		Rotation (°)	
Method	Mean	Median	Mean	Median
P15	3.762	2.771	1.772	0.857
CV	12.943	1.444	9.675	0.175
CA	22.413	0.634	10.114	0.170

Table S16 reports both mean and median errors. P15 has lower means but higher pooled medians than either extrapolator: the baselines combine many small errors with a large-error tail. Using each state's mean over eight horizons, P15 has lower translation error than CV and CA on 42/84 and 37/84 states, and lower rotation error on 27/84 and 34/84. Thus the aggregate means do not establish uniform prediction superiority. The comparison characterizes offline error distributions under GT motion inputs.

G.4. Closed-Loop Alignment

The planned comparison will align observations, predictions, actual futures, action application, and contact in physical time. It will include P05's adverse sphere condition, a sparse-trajectory condition, and a matched future-to-action intervention, recording scene/camera identity, units, visibility, outcomes, and selection rules. Synchronized rollout exports, real-robot and forecast-intervention videos are **[pending]**.

H. Additional Training and Execution Studies

H.1. Paired Training-Recipe Study

The recipe study tests two development combinations, G (frozen DINOv3-S) and I (trainable CNN), both with P07 supervision, the full visual window, and current-target RGB reconstruction. For each profile, training seeds 0 and 1 compare global batch 16 on sixteen GPUs against batch 64 on thirty-two GPUs. The larger-batch recipe doubles the learning rate from 2×10^{-4} to 4×10^{-4} and reduces warm-up from 1,000 to 250 updates, preserving 16,000 warm-up samples. This is a joint recipe and resource comparison, not an isolated batch-size intervention.

All eight runs complete four epochs and 641,664 samples. B16 uses 40,104 updates; B64 uses 10,028, retaining each epoch's 32-sample tail without padding, repetition, or dropping. AdamW, clipping, FP32/TF32, losses, normalization, and the four-epoch cosine exposure are held fixed. All final checkpoints undergo the same sixteen-device full350 validation and the original 1,400-scene evaluation with paired scene and inference seeds. The seed-0 B16 runs are fresh reruns: their scene-level success outcomes exactly reproduce the corresponding BASE2 endpoints. They do not add a new independent training seed. Seed 1 supplies the additional realization.

Performance and variability. For G, B64 minus B16 gives $+0.43$ and -0.86 percentage points at seeds 0 and 1, with paired 95% intervals $[-1.21, 2.07]$ and $[-2.50, 0.79]$. For I, the changes reverse direction: -3.64 $[-5.64, -1.64]$ and $+5.57$ $[3.57, 7.57]$ points. At B16, I falls from 935 to 857 successes between seeds; phase passage falls from 72 to 14. Neither the original seed-0 score nor the mean response to a recipe change establishes stability. The matched E/F controls have not been repeated at seed 1.

The two-seed mean recipe differences are -0.21 $[-1.36, 0.96]$ points for G and $+0.96$ $[-0.50, 2.39]$ for I. Intervals use 20,000 percentile bootstrap samples, RNG seed 20261017, resampling 100 scene identities within each task and retaining their batch and training-seed pairing. They condition on these two trained seeds and fourteen tasks; they do not estimate variation over new training seeds. Table S18 also shows task losses hidden by aggregate scores: G loses 17 XYZ successes at seed 0 and 15 SO(3) successes at seed 1. Auxiliary metrics are not uniformly preserved; for example, I's pose rotation error increases by 6.17% and 14.35%.

Time and adoption. Mean job-creation-to-result time decreases from 302.43 to 120.77 minutes for G and from 257.81 to 101.38 minutes for I. These measured times include queueing, setup, training, saving, validation, export, and evaluation. They do not identify a hardware-only speedup. Training-compute GPU-hours decrease from 70.22 to 52.47 for G and from 59.42 to 42.31 for I; this measure excludes saving and validation and is not a billing record. The predeclared working tolerance was a three-point aggregate SR loss. G satisfies it for both observed seeds, whereas I does not. The project therefore retains B16 as the general recipe; B64 is a conditional option for this G configuration when aggregate SR is the criterion. It is not adopted for I, task-sensitive comparisons, or an untested final design.

H.2. Frozen Three-State Gripper Readout

The execution study freezes the fifth-epoch P03 policy and learns a three-state readout. Its 889-dimensional input concatenates the current action hidden state at the final Euler evaluation ($\tau = 0.8$), eight frames of available end-effector pose, gripper opening and validity, the sampled six-dimensional arm-action horizon, and the current raw gripper value. A 128-hidden-unit GELU MLP predicts open/hold/close logits; argmax replaces only the executed

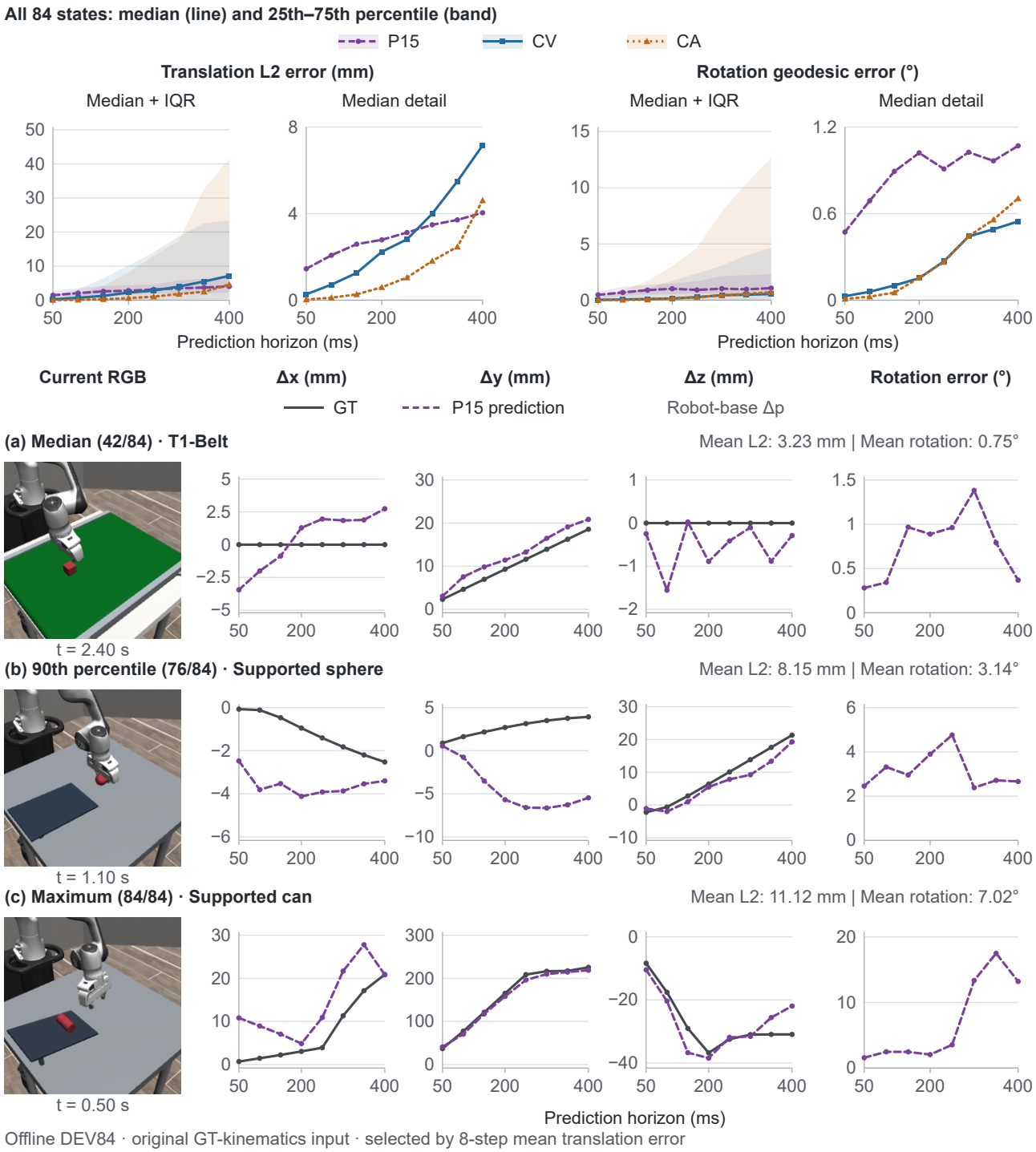


Figure S4. Offline P15 predictions and kinematic extrapolation. The DEV84 probe set contains 84 query states from 28 episodes across fourteen tasks. Top: P15, constant-velocity (CV), and constant-acceleration (CA) median translation L2 and rotation geodesic errors at each horizon; bands show the 25th–75th percentiles across states, not confidence intervals. The adjacent Median detail panels show the same median curves on expanded linear scales, without IQR bands. Bottom: we sort states by their mean translation L2 error over eight horizons and select nearest ranks 42, 76, and 84 (50th, 90th, and 100th percentiles), breaking ties by state index. Each row shows the original current image, predicted and true cumulative translation in robot-base coordinates, and rotation geodesic error at 50–400 ms. The model retains its GT target kinematics and robot-state history. These diagnostic examples characterize offline prediction errors; they do not measure closed-loop contact or task success.

Table S17. Completed training-recipe study, four epochs per run. G uses frozen DINOv3-S and I a trainable CNN; both use the corresponding P07 combination. Batch 64 jointly changes learning rate and device count (details in text). Training time excludes saving and validation. Job time runs from job creation through complete evaluation, including the observed queue and startup.

Profile	Seed	Batch	GPUs	Success /1,400	SR (%)	Train (min)	Job (min)
G	0	16	16	849	60.64	263.40	301.28
G	0	64	32	855	61.07	99.27	121.32
G	1	16	16	834	59.57	263.28	303.58
G	1	64	32	822	58.71	97.48	120.23
I	0	16	16	935	66.79	220.75	257.20
I	0	64	32	884	63.14	78.86	101.74
I	1	16	16	857	61.21	224.91	258.42
I	1	64	32	935	66.79	79.80	101.02

Table S18. Training-recipe successes per task (100 scenes per entry). B16 and B64 include their respective learning-rate/device settings; s0/s1 denote training seeds. All runs use the same accepted scene identities and inference seeds. Differences across tasks remain visible despite similar totals.

	G: frozen DINOv3-S				I: trainable CNN			
Task	B16/s0	B64/s0	B16/s1	B64/s1	B16/s0	B64/s0	B16/s1	B64/s1
Grasp	92	87	91	89	90	95	87	96
Belt	100	100	100	97	97	100	99	100
Occlusion	100	99	100	100	97	98	97	100
SO(3)	87	88	68	53	86	75	56	55
XYZ	75	58	60	69	82	69	77	85
SE(3) placement	0	0	0	0	0	0	1	0
Trans. placement	1	10	10	4	4	5	13	29
Full passage	0	0	0	0	0	0	0	0
Phase passage	0	0	13	0	72	32	14	46
Can	99	99	97	98	100	98	99	100
Slider	83	81	76	80	81	86	75	89
Sphere	40	50	50	57	56	47	60	73
Tabletop sphere	100	99	99	100	98	99	99	100
Replan	72	84	70	75	72	80	80	62
Total	849	855	834	822	935	884	857	935

slot-0 gripper command. A second head with the same hidden width reads only the raw gripper value. No contact truth or future label is added to the observation. The frozen policy, normalization, five Euler steps, and synchronous single-step execution remain common.

Supervision and offline evaluation. The original 11,200 training and 350 validation demonstrations supply 1,324,374 and 41,533 valid issued-command labels; training and validation source seeds are disjoint within each task. Every valid current command is retained, including windows whose later action slots are invalid. Only capture demonstrations contain hold labels: 2,562 in training (0.1934% of all labels) and 79 in validation. The other thirteen tasks have no valid hold labels; all phase-passage training and validation labels are close. This limits what the decoder can learn about useful holds outside capture.

Both heads use seed 0, twenty epochs, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}, batch 4,096, gradient clipping at 1, and ordinary cross-entropy without class reweighting. Feature means and variances use training data only. Each head is selected by the lowest validation mean of the three per-class NLLs, choosing the earliest epoch in a tie. Context selects epoch 15 with NLL 0.01330; raw-only selects epoch 20 with NLL 0.19640.

The context head recalls 79/79 hold labels with precision 79/83, versus 75/79 and 75/85 for raw-only (Table S19). It correctly classifies 137/153 one- or two-tick pulses, compared with 93/153; neither head incorrectly maps these pulses to hold. These counts describe frozen features on demonstration observations, not closed-loop policy perfor-

Table S19. Frozen-decoder validation confusion matrices: 41,533 valid issued commands from 350 demonstrations. Rows are ground truth; columns are predicted open/hold/close. These are offline classification counts, not policy successes.

	Raw only			Context		
GT	Open	Hold	Close	Open	Hold	Close
Open	8003	1	214	8132	0	86
Hold	0	75	4	0	79	0
Close	135	9	33092	105	4	33127

mance. All three per-class NLLs improve. The 79 hold labels receive one third of the checkpoint-selection metric and account for 74.5% of its reduction; training cross-entropy remains unweighted. The context head still makes 86 open-to-close and 105 close-to-open validation errors, so the mean loss does not constrain the cost of an individual closed-loop decision.

Predefined regression cases. The context head is tested on eleven known development cases. It succeeds on E5 capture scenes 0/9/14/80/85, E5 replan scenes 49/88, and E7 XYZ scene 64 and replan scene 7; the E7 cases reuse the E5-trained head without further fitting. It fails on E5 capture scene 52 and replan scene 35. Thus 9/11 pass the regression list, below the predefined all-eleven requirement. The planned 140 new scenes with three decoders (420 rollouts) are not run, and the learned head is not adopted. This case count is not an independent success-rate estimate.

Both public action APIs preserve the original outputs and RNG in observation mode. Twenty-eight sign/dead-zone preflight trials reproduce their original commands and outcomes. Host recomputation checks 608 issued commands: maximum CPU-logit error is 3.82×10^{-6} and argmax commands agree exactly. The other six action dimensions and predicted horizons also pass the recorded comparisons. Complete per-frame policy-input tensors were not saved, so this is not a full input-tensor equality check.

Physical failure sequence. Figure S5 shows both failed cases. In each, the first intervention reverses an approximately -1 open command into $+1$ close; it is not merely suppression of a near-zero output. Capture changes at 0.650 s, followed by opening divergence at 0.666 s and no bilateral contact. Replan applies the altered command at 1.550 s, after its native one-tick queue; opening diverges at 1.556 s. The learned head fails at 1.324 s (tray contact) and 5.174 s (downstream exit), respectively. Sign succeeds in both; the fixed dead zone succeeds in replan but also fails in capture. Recorded predictions, commands, and 2 ms physical fields match their references before intervention. Subsequent arm and object divergence reflects closed-loop feedback. These records establish failures of this readout under the recorded conditions; they do not test a future-to-action prediction interface or establish decoder performance on unseen scenes.

Retained-feature diagnostics. At the first reversal in capture 52 and replan 35, replacing only the action hidden state with the preceding frame's value changes the close-minus-open logit margin from $+2.452/+0.700$ to $-13.839/-9.811$. Conversely, inserting the current hidden state into the preceding feature vector yields $+2.243/+3.184$; replacing only the raw value does not remove either reversal. These substitutions measure a local response of the frozen classifier to mixed features, which need not correspond to realizable observations or successful executions. The first reversal precedes any prior intervention by this head. The retained data do not separate demonstration-to-policy feature mismatch from sampling sensitivity of the final Euler hidden state; both explanations remain untested. The failed regression result is unchanged.

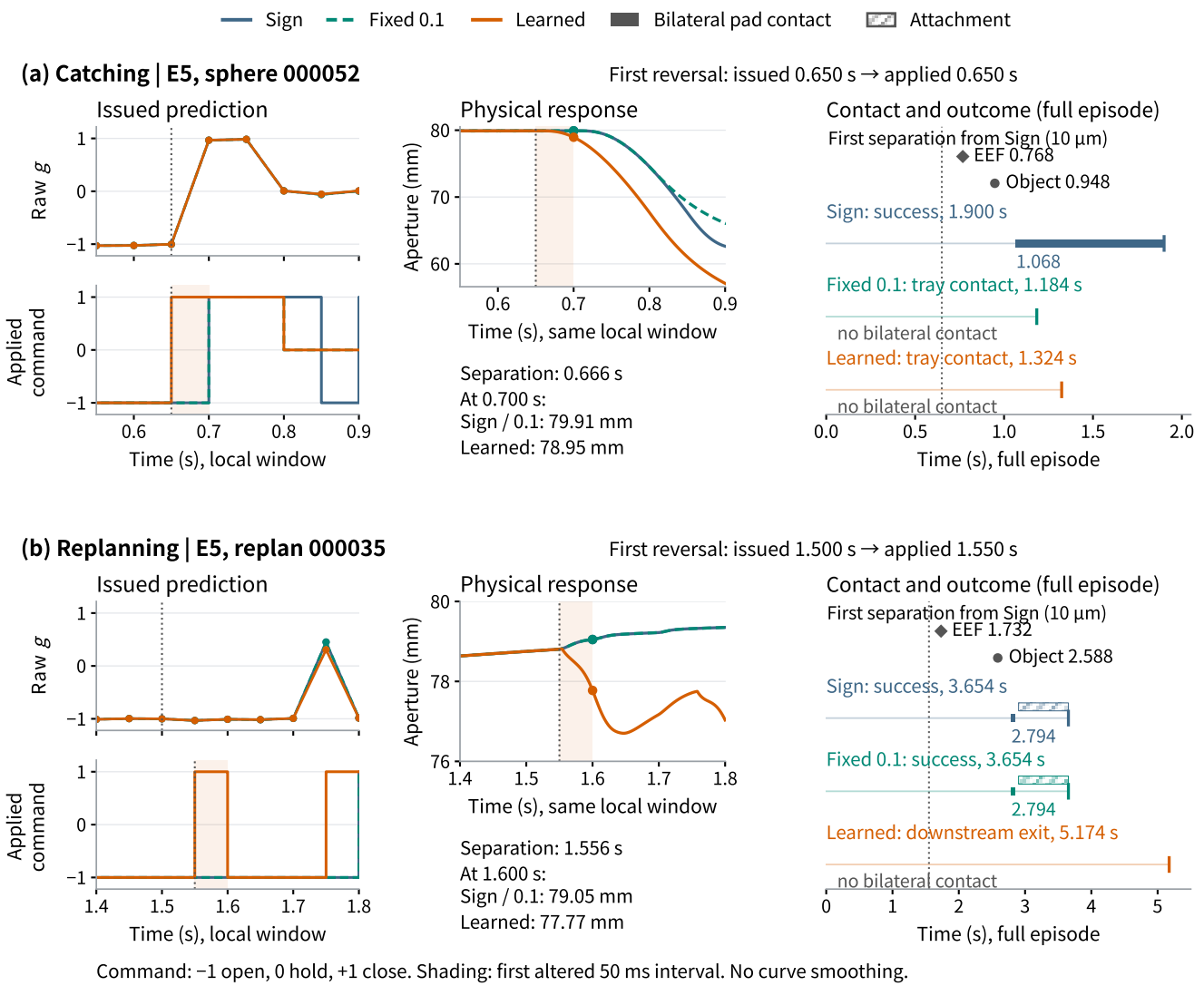


Figure S5. Frozen gripper-readout failures on the two failed predefined regression cases. Commands and measured opening use physical time; event strips retain each run's contact and terminal outcome. Sign, the fixed 0.1 dead zone, and the learned context head use the same frozen policy. The head's first intervention reverses opening into closing, before the observed physical divergence. Detailed windows and full-run events serve different time ranges, labeled on their axes. Contact refers to actual bilateral collision; replan attachment is a separate event. Raw predictions are sampled at 20 Hz and physical states every 2 ms. Separation of the learned run from Sign uses a 10 μm aperture or componentwise position threshold. The cases are selected development diagnostics, not new-scene success estimates or future-prediction interventions.